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AR Mount Guide

BSc (Hons) Computer Science and Software Engineering

Undergraduate Thesis Report

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Abstract

The following research paper focuses on enhancing mountain exploration for tourists and regular people by utilizing modern technology such as Augmented Reality (AR). AR is gaining more popularity due its immersive user experience where user can view any objects in real scenario. With tourism increasing in Nepal, mountain exploration can be quite difficult for tourists due to its harsh climatic conditions and lack of knowledge regarding surrounding landscapes. Conventional methods, including books and internet searches, might be challenging to learn about the nearby mountains. In order to address that issue, the following research focuses on how the AR Mount Guide project is a revolutionary interactive mobile application that utilizes augmented reality features to enhance user experience in their way of viewing mountains. The study also focuses on determining user's orientation in real time using geolocation, gyroscope and compass data. Furthermore, the study aims to improve user experience by offering an indoor mountain simulation that allows users to see mountains in a private space. The project aims to provide an interactive AR experience by allowing users to scan the surroundings and get the mountains located in that particular direction along with their detailed information.

Keywords: Mountain Identification, Augmented Reality, AR Tourism, Geospatial tech, AR with Geospatial tech, 3D Model with AR

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BSC.CS & SE

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List of abbreviation

AR- Augmented Reality

GPS - Global Positioning System

GIS - Geographic Information System

UI – User Interface

UX- User Experience

API- Application Programming Interface

SDK- Software Development Kit

3D- Three dimensional

DB- Database

Mb- Megabyte

RAM- Random Access Memory

OS- Operating System,

VS Code- Visual Studio Code

AR Core- Augmented Reality Core

DB- Database

API- Application Programming Interface

UI/UX- User Interface and User Experience

Chapter 1: Introduction

1.1 Background

Nepal is seeing a rise in tourism as more and more tourists travel there for trekking and hiking. Despite the large number of visitors to Nepal, not all of them are likely to be familiar with the country's mountains, landscapes, and other geographical aspects. The geographical features of Nepal are not fully understood by even an ordinary person. In addition, individuals frequently have to rely on maps, guidebooks, and search engines. Despite their need, these traditional approaches do not offer the user an interactive, clear, and intuitive experience.

Similar to this, augmented reality is growing in popularity over time as it improves the user experience by overlaying virtual elements with the actual environment in the user's field of vision (Syed, 2023) . Furthermore, utilizing contemporary technologies like augmented reality in mobile apps has several advantages as it enables users to see any items that improve the user experience (Davidaviciene, 2021). Additionally, augmented reality enhances navigation and tourism by enabling users to access real-time interactive data and information about any geographical characteristics when combined with geospatial technology (Manggopa, 2022). Combining augmented reality with tourism would be very beneficial in producing immersive and distinctive experiences that draw tourists and highlight the value of natural and cultural heritages (Georgios Prapas, 2023).

To concisely explain, the following research aims to utilize the full potential of AR overlays for a user immersive experience of viewing mountains.

1.2 Problem Statement

In the present scenario, people often rely on conventional methods such as tour guides, guided books, or any online searches to know about specific mountains. While these methods are necessary, they require a lot of manual effort, where it may be time-consuming, and in some cases, the provided information is not clear and intuitive (Ricart, 2023). Since Nepal is a naturally gifted country with many snow-capped mountain peaks, it is difficult for every individual to know about particular peaks.

Furthermore, there are comparatively limited options for those who cannot physically visit mountain peaks and sceneries to experience the mountains or even learn anything about them. They have limited resources and obstacles that make mountain exploration challenging. Additionally, one of the major problems is in the domain of education, where students are taught using static images of mountains, which hinders their ability to interact with the content being taught. Students struggle to learn about geographical characteristics, terrain structures, and other mountain regions because of their lack of visualization.

1.3 Proposed Solution

The AR Mount Guide project is a revolutionary mobile application that combines modern technologies such as augmented reality overlay along with geospatial technology for an immersive experience for viewing mountains (Manggopa, 2022). The primary goal is to provide an interactive AR experience where the user scans their surroundings and gets the mountain located in the particular direction. Additionally, the application offers interior simulation, allowing the user to view mountains in any direction by scanning any side of their room.

In addition, the AR mount guide will give those with physical limitations and impairments an engaging and entertaining method of exploring mountains. In addition, students can observe 3D models of mountain peaks and real-time overlays of mountain terrain with geographical information. This enhances the teaching and learning process in general, making it more interesting, enjoyable, and approachable.

1.4 Aims and Objectives

The primary aim of this study is to research how modern technology like augmented reality (AR) can enhance mountain exploration for tourists and for regular people. It aims to increase the overall experience of users by creating an easy-to-use, cost-efficient, and more user-friendly AR mobile application.

The main objectives of this study are as follows:

- To research and implement a fully functional AR mobile application for overlaying the mountain and its information.

- To enhance an immersive and enjoyable experience by providing indoor AR simulation.
- To provide an optimized 3D model display of mountains with its detailed information.
- To study geolocation, gyroscope, and compass data for real-time user data.

1.5 Feasibility Study

This chapter's supplied section focuses on the feasibility, practicality, and feasible nature of the suggested study (DRURY, 2025). In addition to determining whether the product can be finished in a reasonable amount of time, it will offer a comprehensive implementation guide for the AR Mount Guide. The viability of the market, operations, finances, and ethics are all examined in this study.

1.5.1 Technical Feasibility

Due to its ease of use and compatibility with both AR and geolocation capabilities, the AR Mount Guide project makes use of Flutter for mobile app development (Team, 2025). Additionally, Node.js and MongoDB will be used for backend technologies since they make handling geographical data and queries simpler (Tarushee Kumar, 2021). According to additional research, Flutter AR plugins make it simpler to integrate AR functionality. Backend technology and mobile app development experience are needed for the next project.

1.5.2 Operational Feasibility

The AR Mount Guide project will have a simple user interface that requires little setup. All types of users will be able to access this. However, there is always a need for guidance for those with less technical expertise. Users will be presented an interactive guide to help them use the software for that purpose (Akash Chakraborty, 2022). This will make this app more viable for a wider range of consumers. In addition, viewers can study the 3D model and obtain a thorough description of the mountain, as well as tour the mountains using indoor simulation.

1.5.3 Financial Feasibility

The cost of utilizing augmented reality features is one of the main challenges (Martins, 2024). The majority of applications that give users access to augmented reality (AR) features are expensive to subscribe to. Nevertheless, AR Mount Guide addresses this problem by offering a user-friendly and inexpensive AR mobile application. The fact that this application is free to use increases its accessibility for users.

1.5.4 Market Feasibility

People are becoming more interested in adopting these kinds of applications again because of the increasing demand for augmented reality and its use in tourism (Team, 2023). Furthermore, the AR mount guide application is a ground-breaking tool that will improve people's overall experience when they explore mountains on their cellphones, as opposed to depending on more conventional approaches like using search engines, books, and a tour guide. As a result, when this application is released onto the market, more and more people will want to use it.

1.6 Intellectual Challenges

To ensure overall positive usability and effectiveness, the presented study consists of a number of intellectual problems that require appropriate attention (*TTConsultants, 2025*). Furthermore, because AR has hardware constraints, incorporating it into mobile apps might be difficult. The following list includes a few of the intellectual difficulties:

1.6.1 Hardware Limitation

The compatibility of AR components with mobile devices makes integrating them into mobile apps one of the most challenging tasks (InnoVirtuoso, 2023). Not all smartphones have been upgraded to support augmented reality. The system's general usability and functionality are impacted by this (Merve Yavuz, 2021). Moreover, 3D models and real-time overlays in AR applications demand a lot of processing resources. Further, most of the mobile devices struggle with object tracking in outdoor lighting conditions.

1.6.2 Environmental and Weather adaptability

The majority of mobile devices that support augmented reality have trouble adjusting to shifting surrounding situations. Rainy weather and fogging can make AR apps

challenging to utilize (Team, 2025). Furthermore, most mobile devices will have trouble using AR features because of the intense sunshine. Network connectivity problems and poor GPS accuracy due to mountainous terrain may arise when trekking and hiking.

1.6.3 Power Consumption

For AR Mount Guide, it requires the usage of a constant camera, the user's current location, and rendering a 3D overlay. Due to this, it consumes most of the battery health of mobile devices. This also causes constant heating of mobile devices (Tong Li, 2022). The constant processing of 3D models and AR overlays makes it challenging for everyone to use this technology.

1.7 Thesis Report Format

Chapters	Title	Topics	Subject Matter
1	Introduction	1.1 Background 1.2 Problem statement 1.3 Proposed Solution 1.4 Aims and Objective 1.5 Feasibility Study 1.5.1 Technical Feasibility 1.5.2 Operational Feasibility 1.2.3 Financial Feasibility 1.2.4 Market Feasibility 1.6 Intellectual Challenges 1.7 Contextual Report Format	Outlines the project's goals, objectives, problems, and suggested solutions in the introduction section. It also discusses the feasibility of the study presented in terms of its technical, operational, financial, and commercial viability. It also offers the intellectual challenges that could arise when this application is being developed.
2	Project Plan	2.1 Gantt chart 2.1.1 Semester 1 Gantt chart 2.1.2 Semester 2 Gantt chart 2.1.3 Revised Semester- 2 Gantt chart 2.2 Risk Analysis 2.3 Workflow Diagram	Offers a thorough Gantt chart for the first and second semesters. detects risks as well and offers ways to mitigate them. also offers the work breakdown structure, which shows the work that has been provided broken down step-by-step..

3	Literature Review (LR)	3.1 Overview 3.2 Review on existing projects 3.3 Key Terms 3.4 Literature Reviews 3.5 Summary of Literature Review 3.6 Justification	Focuses on researching the articles and journal relevant to the given project. Covers literature review of all article that has been utilized along with the overall summary of literature review.
4	Primary Research	4.1 Objectives of primary research 4.2 Survey Questionnaire 4.2.1 Survey Responses 4.3 Data Collection and Analysis 4.3.1 Data Analysis	Focuses on main aim of doing primary research, survey questionnaire. Also focuses on detailed data collection and analysis.
5	Artifact Planning	5.1 Requirement Analysis 5.1.1 Functional Requirements 5.1.2 Non-Functional Requirements 5.1.3 Usability Requirements 5.2 Software Requirements 5.3 Hardware Requirements 5.3.1 Development Requirements 5.3.2 Deployment Requirements 5.4 System Design 5.4.1 Use Case Diagram 5.4.2 Database Design 5.4.3 Activity Diagram	Focuses on functional, non- functional and usability requirements required for developing the software. Also focuses on software and hardware requirements that will be used for development of the project. Similarly, it focuses on system design that includes use case diagram and database design.
6	Testing and Evaluation plan	6.1 Testing Plan 6.2 Evaluation Plan	Focuses on testing strategy as well as evaluation strategy
7	Development Process	7.1 Methodology 7.2 Tech stack utilized 7.3 Development Planning and data collection	Discuss various phases of Ar Mount guide development, focusing on methodology, problems, features and integration of ar in the application

		7.4 Managing android permission and project initialization 7.5 ArCore integration 7.6 Ar Rendering and Mountain Overlay 7.7 Frontend Display 7.8 Backend Integration and Interoperability	
8	Testing	8.1 Unit testing 8.1.1 Unit test case scenario for register 8.1.2 Unit test case scenario for login 8.1.3 Unit test case scenario for Ar mountain overlay 8.2 System Testing 8.3 Integrated Testing	Contains all unit test, system test and integrated test conduction after development of the application
9	Evaluation	9.1 User Evaluation 9.2 Performance Analysis	Explain user evaluation and feedback as well as explaining the performance
10	Critical Analysis	Critical analysis	Critically analyse the project overall weakness, strengths.
11	Limitation and Future Scope	11.1 Limitation 11.2 Future Scope	Describes limitations and possible improvements
12	Conclusion	Conclusion	Overall conclusion of this research
13	Reference	References	Includes citation in this report
14	Appendices	14.1 Weekly Progress report for Undergraduate project 14.2 Proposal for RMET 14.3 Ethics form 14.4 Testing figure 14.5 Google form 14.6 Poster	Contains necessary images

Table 1: Thesis Report Format

Chapter 2: Project Plan

2.1 Gantt chart

2.1.1 Semester 1 Gantt chart

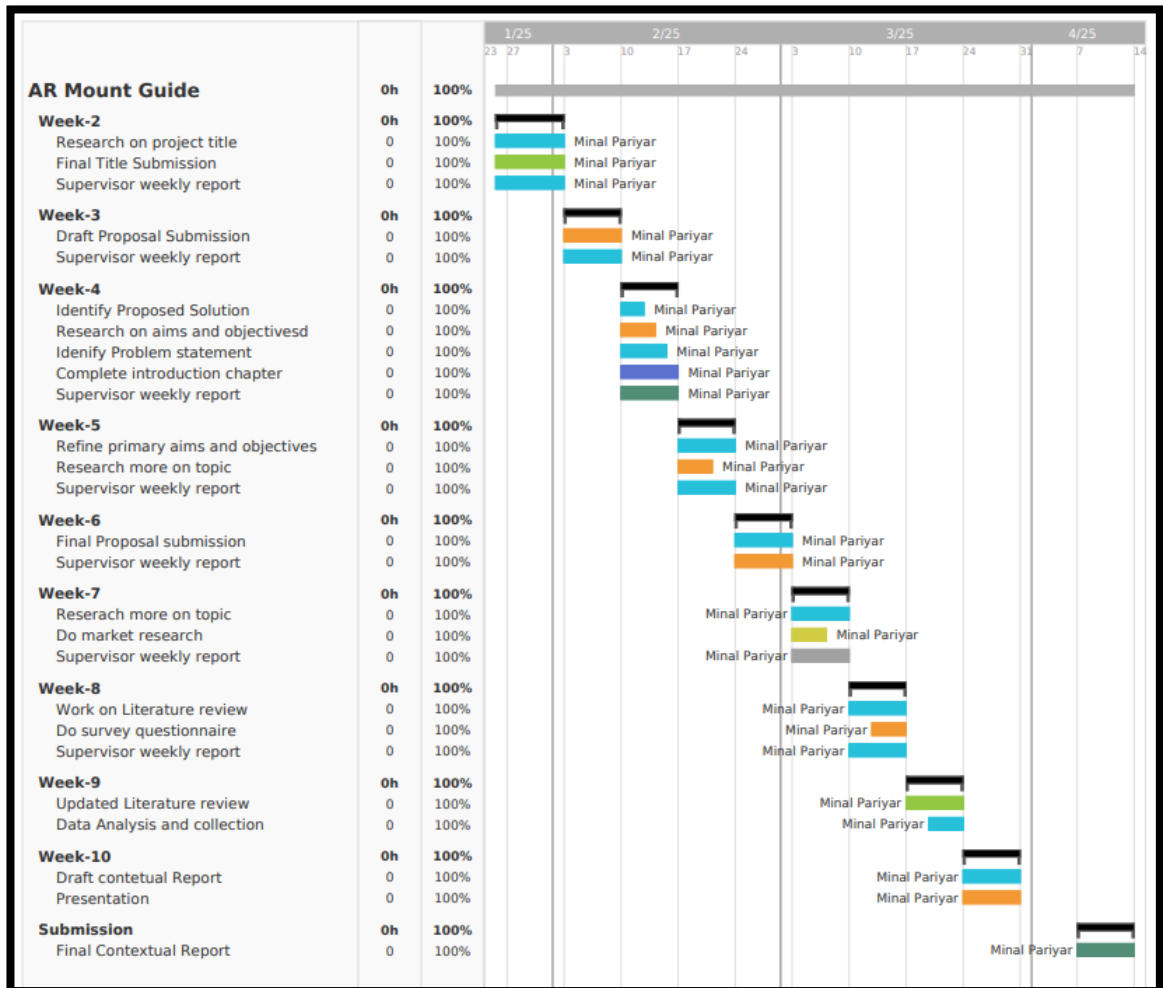


Figure 1: Semester 1 Gantt chart

2.1.2 Semester 2 Gantt chart

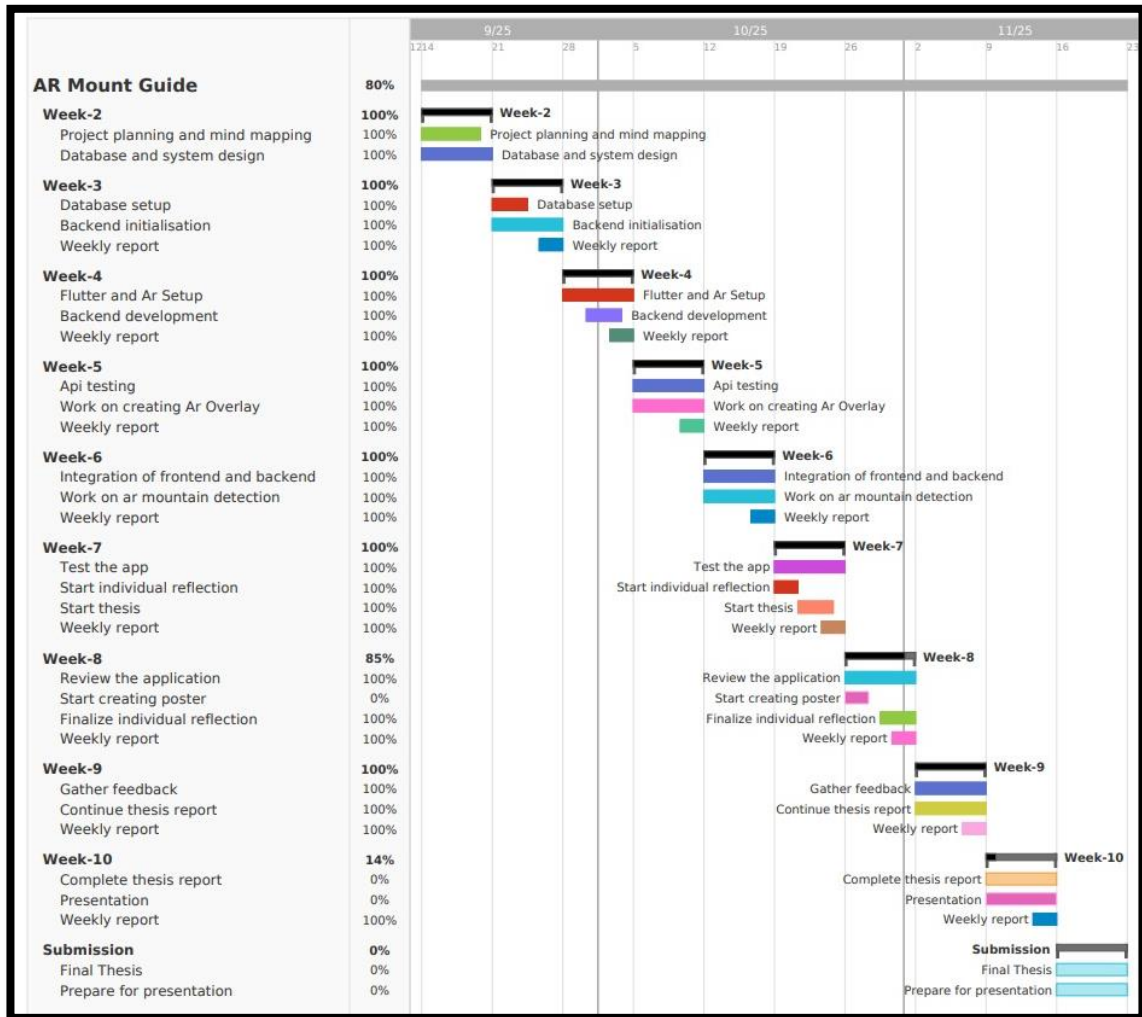


Figure 2: Revised Semester- 2 Gantt chart

2.2 Risk Analysis

One of the most crucial aspects of developing an application is risk analysis, which examines any hazards that might have a detrimental effect on projects (Yasar, 2021). Similar to this, there may be certain risks or uncertainties associated with the AR Mount Guide project that might have an impact on the project as a whole. Through risk analysis, it can employ a number of mitigation strategies to lessen the adverse impact. Accordingly, the risk assessment that follows offers a list of potential vulnerabilities that might arise throughout the project's development and implementation (GLADREY, 2025).

Possible Risks	Impact	Likelihood	Mitigation Techniques
Inaccurate Orientation and placement	HIGH	LIKELY	Implement alternative technique for detecting location
Environmental Constraints	HIGH	POSSIBLE	Notify users regarding bad weather condition and relocate in an area with good lighting
Hardware Limitation	HIGH	MODERATE	Use a lightweight augmented reality engine to optimize the app.
Poor internet Connectivity	LOW	LOW	Implement offline mode
High Battery Consumption	MEDIUM	MODERATE	Implement power saving technique for proper optimization
Failure of AR in poor lighting	MEDIUM	MODERATE	Inform users to relocate to an area with adequate lighting.
Limited Storage Capacity	HIGH	HIGH	Store 3d model and other assets in cloud platform like cloundinary

Table 2: Risk Analysis

2.3 Workflow Diagram

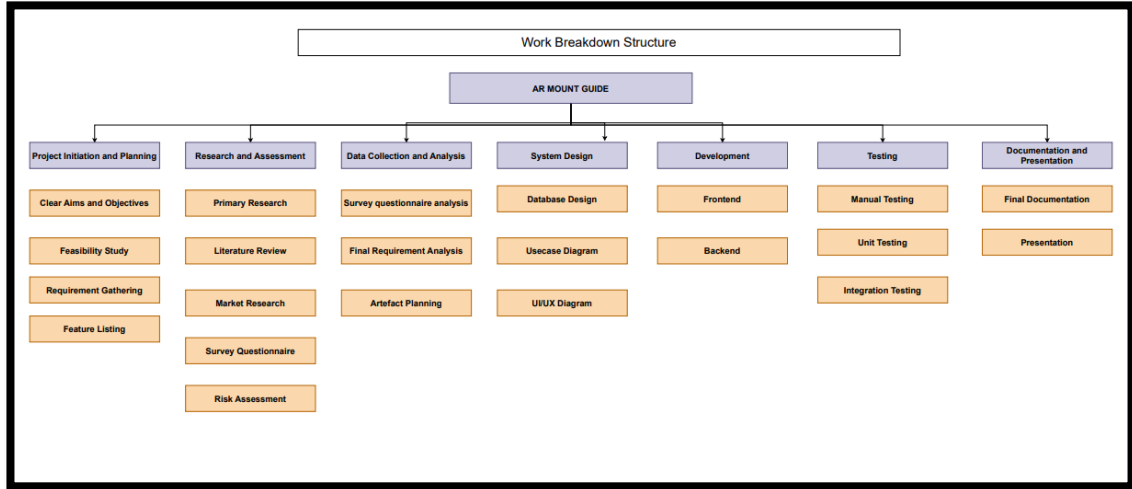


Figure 3 : Workflow Diagram

Chapter 3: Literature Review (LR)

3.1 Overview

A literature review is a comprehensive summary of previous scholarly research papers that aids in the development of familiarity and approach for any given research topic (McCombes, 2023). In Similar manger, the following section focuses on analysis of existing research paper for Ar Mount Guide. The AR Mount Guide project is a revolutionary and interactive mobile application that utilizes augmented reality features along with geospatial technology to enhance user experience in their way of viewing mountains. The main objective of this research paper is how, modern technology such as AR can enhance the user experience of people when viewing mountains.

3.2 Review of existing projects

3.2.1 AR Alpine Guide

Ar Alpine Guide is an innovative mobile application that leverages the usage of AI and geolocation technologies to identify peaks as well as hillsides. The project covers 1,000,000 peaks worldwide; maps can be downloaded for offline use; sunrise and sunset times for any peak; 3D mode for a realistic view; 360 panoramic views; and, most importantly, it is free to use. This project is used by a large number of people worldwide.

But the research that is offered will only look at ten mountains. A mountain is also simulated indoors using augmented reality in the AR Mount Guide app.

3.2.2 Peak Finder

The offered peak finder app is a cross-platform smartphone application that facilitates the use of augmented reality to locate nearby hills and mountain peaks. In a similar vein, the following offline app features 1,000,000 mountain summits worldwide. In a similar vein, the app allows the user to label mountains and offers a panoramic view (Fabio Soldati, 2021). Additionally, in locations with weak GPS signals, the precision of the compass and GPS may be impacted. Likewise, this program might not work on some older devices. Additionally, a paid membership was needed to use the program.

3.2.3 Peak Visor

The Peak Visor application, which is offered, is intended to encourage people to explore more. The following app helps users plan journeys efficiently by offering 3D topographic maps with comprehensive hiking pathways. A cutting-edge technology 3D compass included into the app precisely determines the height of every mountain peak that appears (team, 2024).

3.3 Key Terms

3.3.1 Augmented Reality (AR)

The term augmented reality refers to the integration of virtual objects into real-world environments. Users get to experience artificial environments in real-world applications (Hayes, 2025).

3.3.2 Geospatial Technology

In general, the term "geospatial technologies" refers to the tools and devices used in mapping and geographic positioning. These methods aid in gathering and examining particular places on Earth (Bomley, 2024).

3.3.3 Geographic Information System (GIS)

A component of geospatial technology, GIS shows data that has been geographically referenced (government, 2023). GIS makes use of location-based data, which can be represented by zip code, address, or latitude and longitude. People can compare the

locations of various objects to learn how they relate to one another with the use of GIS (team, 2025).

3.3.4 Global Positioning System (GPS)

Another geospatial technology tool is GPS, which uses satellites to provide the user's present location as well as extremely accurate positioning of any location (Novatel, 2025). For example, the user can utilize GPS to access their present location.

3.3.5 Geolocation

Geolocation technology uses GPS, Wi-Fi, or cellular networks to determine a user's present location, including their address, latitude, and longitude (team, 2021). It tracks specific latitude and longitude coordinates using a positioning system.

3.3.6 3D Modelling

3D modeling is the process of utilizing specialized software to create digital representations of three-dimensional objects and shapes. Likewise, a 3D model is a digital representation of a 3D object (team, 2023).

3.3.7 AR Overlay

Generally speaking, "AR overlay" refers to the process of using your mobile devices to scan and display virtual things in a real-world setting. In a similar vein, AR overlays improve location-based services' accuracy and usefulness (Serrano, 2024).

3.3.8 Camera Sensor Integration

Computer vision, industrial applications, augmented reality, and many more fields use camera sensor integration, which is the process of integrating camera devices to take pictures or movies (Arora, 2021).

3.3.9 Compass

The device that gives us the cardinal directions—north, south, east, and west—is a compass. We can determine where we are going with the use of a compass. Without a compass to indicate the correct direction, even a map is useless (apple, 2025).

3.3.10 Gyroscope

The device that maintains direction despite movement is a gyroscope, which makes use of the idea of angular momentum. In a similar vein, smartphones employ a gyroscope to identify the orientation of the screen.

3.3.11 Object Detection

One of the most important technologies in many domains, including medical robotics, is object detection, which identifies the primary object among several objects in a single frame (Ph.D, 2024).

3.3.11 Outdoor Navigation

The term "outdoor navigation" describes navigating through outdoor spaces like parks, roads, hiking routes, and mountainous regions. GPS technology is typically used in outdoor navigation to help with location determination and navigation (team, 2025).

3.3.12 Indoor Navigation

In general, indoor navigation relates to assisting users in navigating within their own environment, such as workplaces, rooms, etc. (Butler, 2024).

3.4 Literature Review

3.4.1 Understanding AR technologies

The research paper proposed by (Sakib, 2022) focuses on detailed overview of augmented reality and how it has impacted in user engagement with AR applications, The study examines how augmented reality (AR) has emerged as the most significant technology that combines the virtual and real worlds to give people immersive experiences. Applications for augmented reality are constantly being developed in a variety of industries, including healthcare, education, retail, and entertainment. The origin of AR and its fundamental elements are also examined in the study.

The three main pillars of augmented reality are real-time rendering, display technology, and tracking and registration. The article also discusses how AR has improved a number of disciplines. For example, employing AR in surgical operations can help doctors and surgeons in the health industry. Additionally, AR based cultural heritage guide has helped in tourism industry. Similarly, augmented reality has made contributions to the medical,

military, tourism, and educational sectors Further, AR is also emerging with technology such as Artificial intelligence, Internet of things and 5G networks. All things considered, the research study that follows is mainly concerned with the development of AR and how it might benefit individuals in many industries (Sakib, 2022).

Due to advancement of AR in tourism, a lot of tourist as well as regular people are finding it comfortable to use AR as it allow visually pleasing engagement to view any geographic location. Reliable tracking system with AR can help in indoor exploration of objects easier. While AR tracking is useful, there are certain challenges, such as the lighting conditions of the room, backdrop contrast and different colour intensities. The entire user experience may be impacted by this. Furthermore, not everyone has a device that can support high-power AR, therefore, employing AR restricts the users. Furthermore, a lot of consumers who are not familiar with AR technology could find it difficult to adjust or feel at ease. To add to that, AR mount guide faces challenges related to weather conditions, lighting conditions and GPS inaccuracy, which impact the performance.

3.4.2 In-Depth Review of Augmented Reality: Tracking Technologies, Development Tools, AR Displays, Collaborative AR, and Security Concerns

The following study, which was released in 2023 by (Toqeer Ali Syed, 2023), offers a thorough analysis of the condition of augmented reality today and discusses a number of ways to make it better. The study examines how augmented reality (AR) is becoming more and more popular in the twenty-first century and how it has been applied in many sectors. The study also emphasises how precise monitoring is essential to augmented reality experiences in order to accurately overlay virtual tracking onto the physical world. Additionally, it covers a variety of tracking techniques, including sensor-based tracking, vision-based tracking, marker-based tracking, marker-less tracking, 3D model tracking, and many more.

Additionally, the article focuses on web-based augmented reality as well as other development platforms and tools including ARKit, ARCore, Unreal, and Unity. Additionally, it covers collaborative augmented reality, in which several users engage in real-time interaction with shared virtual information. All things considered, the study

paper that follows provides a thorough analysis of the state of augmented reality today, stressing both its developments and difficulties.

This article's primary focus is on AR tracking technologies. In AR Mount Guide, methods such as model tracking and three-dimensional tracking might be helpful for creating an immersive augmented reality experience. For the creation of user-interactive mobile applications, the availability of powerful AR development kits such as ARKit and ARCore can be helpful. Because collaborative AR enables real-time interactions between numerous users, it has the potential to revolutionize the field.

Despite AR's enormous popularity, there are still several constraints and limitations. Environmental limitation is one of the few. While various tracking techniques are available, these techniques can still fail with less accuracy in the context of real environments, such as low light conditions. In a similar vein, collaborative AR can also be quite difficult to manage due to its complexity. When there are multiple users involved, then there is high complexity in managing real-time interactions.

3.4.3 Evaluation of user experience in Augmented Reality

The following research study conducted by (Vida DAVIDAVIČIENĖ, 2021), is on assessing the user experience of augmented reality (AR), a new and emerging technology. This study also clarifies the ways in which augmented reality has linked the virtual and physical worlds. In addition to technology advancements and globalisation, augmented reality has become increasingly popular due to shifting consumer demands. According to the research, augmented reality (AR) in mobile applications has been studied across a range of businesses and how it may affect how customers see items.

The study main aim is to evaluate user experience and provide recommendation for their development to address this type of issue. Here, a two case study was conducted to evaluate the user experience of augmented reality commercial mobile applications. Two AR application that were active during this reseach such as Arilyn(app that present visual story of product by Kalnapilis) and Inkhunter(app that tests virtual tattoos using augmented reality). Additionally, a number of standardised instruments, including the System Usability Scale, USE, HARUS, ISO usability questionnaire, and User Experience Questionnaire, were used to assess user experience in AR applications. The User Experience Questionnaire was chosen as one of these instruments to assess the user

experience in augmented reality apps as it reflects the overall user experience of object being evaluated.

The evaluation result of Inkhunter an Ariklyn mobile app has shown overall user experience to be positive. Overall, the research conducted by (Vida DAVIDAVIČIENĖ, 2021) identifies key factors such as ease of user interactivity has positively influence user experience in Ar Mobile applications.

With only 30 replies, the sample size for the study article that follows is small. The brief usage length of the participants who were given this software for testing also made it impossible to evaluate long-term usability factors like memory and battery usage. Only two AR-based mobile applications were the subject of additional study, which made it challenging to evaluate the entire AR experience.

3.4.4 Augmented Reality for Geospatial in E- tourism

In many facets of life, including health, education, entertainment, and travel, augmented reality has seen tremendous growth. The following research carried out by (Hiskia Kamang Manggopa, 2022) discussed the implementation of augmented reality in e-tourism and presented various innovations of augmented reality in tourism. According to the author, augmented reality can improve the tourism sector if it is maximised properly. Further integration of AR into e-tourism is one of the major focuses of this research.

This study's primary goal is to examine and pinpoint data sets, research patterns, and methodologies that have been applied since 2017 as well as potential future paths for augmented reality's application in e-tourism. The author describes how few studies have examined the use of augmented reality in the travel industry. But according to the author, augmented reality has a lot to offer the travel industry. All things considered, the study paper that follows gives a summary of current research trends and the future course of augmented reality in e-tourism.

Although augmented reality has the potential to revolutionise the tourist industry, many consumers may find AR apps difficult to use, which restricts their options. Additionally, AR apps in the travel industry gather a lot of location-based data, which increases security and privacy concerns. Furthermore, inaccurate or out-of-date information in AR apps might deceive travellers and undermine consumer confidence.

3.4.5 Augmented Reality Interactive Guide System and Method for Tourist Attractions Based on Geographic Location

The study conducted by (Xueqiu Zhuang, 2022) aims to study an AR-based interactive guide system that provides tourism services for tourists in tourist attractions based on geographic location. The study also examines current GPS technology and suggests an enhanced probabilistic algorithm model for precise visitor location. Applications for smart tourism have greatly evolved thanks to augmented reality and GPS technology. The author briefly describes how combining augmented reality (AR) with location services improves the user experience by giving travellers access to real-time data.

The author focuses on the potential of personalised self-guided tours based on AR and geospatial technology. (Xueqiu Zhuang, 2022) proposes an idea of a mobile application that allows visitors and tourists who are travelling to have their own personalised tours based on their preferences. Furthermore, he also explains how AR can visualise historical monuments and create scenic views of any place.

Utilising AR overlay trail markers, waypoints can provide seamless viewing of mountain terrains. AR can help in interactive learning by providing an overlay of mountains along with its description, which enhances the education. Even while GPS and AR technologies have advanced, many mobile devices still face environmental limitations including signal blockages and weather disruptions. Since the present homographic matrix computation needs to be improved for better spatial alignment, the precision may not be useful for the AR mount guidance app.

3.4.6 Visualising Location-Based Information using Augmented Reality

(Zhang, 2023) explores the use of augmented reality to visualise location-based data in a narrative setting in the research article that follows. The author goes on to explain how augmented reality has advanced and changed the technology in the area of location-based apps. Additionally, four mobile application prototypes that investigate narrative navigation have been examined by the author of this study work.

Four ideas on the idea of narrative navigation are the main subject of the following thesis, which was completed in 2023. Additionally, the study investigates the idea of data

visualisations, which is crucial for improving users' on-site experiences. Overall, the author of this study article aims to demonstrate how storytelling is an effective information-transmission tool that offers special chances to lead people via location-based tales when paired with augmented reality and location-based services.

Although a lot of study has been done on location-based storytelling and augmented reality navigation, little is known about how these two technologies interact to create narrative storytelling. AR overlay can help regular people, trekkers and hikers to visualize mountains. AR mount guide project can benefit from location based service which is essential in calculating distance between users and mountain terrains

While location-based service applications are useful, there is major concern regarding the security and privacy issues. Further, overuse of high consumption of batteries can drain the mobile app and can affect the user experience. Further, not all mobile phones support AR, which limits the users for AR applications.

3.4.7 Intention to use Mobile augmented reality in tourism sector

The following research paper conducted by (Malik Khlaif Gharaibeh, 2021), examines and explore the usage of Mobile Augmented Reality in Tourism. Here, the author describes how augmented reality (AR) has become a visual aid that makes customers more interactive. Performance, expectancy, effort expectancy, social influence, hedonic motivation, facilitating conditions, and price value are the six independent variables that are used in the following study to examine this topic using the framework of the Unified Theory of Acceptance and Use of Technology. The author anticipates a result about the purpose of mobile augmented reality users in the travel industry based on the combination of these criteria.

The first of the many limitations of this research report is that it was only done in one country, which restricts the findings in other countries. Additionally, the report heavily relies on self-reported information gathered via surveys. Although the study report makes a significant contribution, it does not examine consumers' sustained involvement with augmented reality mobile apps.

3.4.8 3D Modeling and Augmented Reality for Learning

The following research paper proposed by (Adel Fridhi, 2023) explores the idea of integration of 3D modeling and augmented reality in civil engineering education. It also explains how AR has capacity to bridge the gap between theoretical knowledge and practical application. Moreover, in civil engineering many student find it difficult to visualize models and design but with emerging technology such as AR, it can help them to interact and visualize various models.

This study article's primary goal is to improve the teaching and learning process by utilising a variety of pedagogical techniques. Students may use AR to learn how to do experiential learning for their projects. In a similar vein, students may collaborate in real time while interacting with virtual items in the actual world using collaborative augmented reality. Furthermore, constructing actual structures in New York City is part of (Adel Fridhi, 2023) study, which strengthens the educational objective.

Moreover, AR when combined with three dimensional model can help in educational purposes for students where they can get practical and experimental visualization of objects. To add to that, Augmented Reality can help in bridge gap between theoretical and practical learning.

The use of AR and 3D models in teaching is revolutionary. However, the setup and maintenance of AR may be expensive. In a similar vein, making a 3D model of an object can take a long time. Additionally, instructors and students may find it challenging to use AR due to technological issues. Furthermore, since not all students have access to devices that support AR, employing AR may restrict its usage.

3.4.9 Augmented Reality Applications for Learning Geography in Primary Education

The following research article issued by (Christina Volioti, 2022) focuses on improving the primary education by combining the augmented reality with geographical features. The research explores on how augmented reality when combined with geography can enhance the learning process and improve user experience and interactivity through proper visualization of geographical places.

The three AR geography apps created for fifth and sixth grades are also reviewed in this article. It also talks about how the three applications have the same architecture and how users were instructed on how to utilise each one. The C# programming language and the Unity game engine were used to create all three applications. The first augmented reality software teaches geography, with a focus on Greece. The second application concentrated on orientation, with the main objective being to help the user comprehend the fundamentals of orientation and how to move about a location. Additionally, the third app seeks to educate people about coastal elements. Overall, AR can enhance geography education by visualising geographical features and interactive maps.

Teachers and pupils may find it challenging to immediately adjust to this kind of technology, which prevents them from utilising the AR application to its fullest. Furthermore, it might be difficult to guarantee that AR content has a high instructional value. Additionally, students may become distracted when using the AR software, which would hinder their ability to master the fundamentals. Additionally, implementing AR might be more expensive and a hurdle for many institutions.

3.4.10 An Indoor Navigation Methodology for Mobile Devices by Integrating Augmented Reality and Semantic Web

(Jesus Ivan Rubio-Sandoval, 2021) research study from 2021 examines the concept of using augmented reality to help people find locations within buildings and learn more about their surroundings. The study also suggests an approach for creating an interior navigation system that combines semantic web technologies with augmented reality.

The research study also proposes an indoor navigation system with aims to support users in finding an office, laboratory, classroom or finding a place according to the staff member. Additionally, in areas with poor GPS signals, these kinds of devices might be employed. Additionally, indoor navigation employs a number of techniques, including pathfinding algorithms, positioning techniques, spatial representation, and augmented reality visualisation. In order to improve the user experience, the system combines AR and semantic web technologies. This includes Dijkstra's algorithms for dynamic route computation, a knowledge graph that stores and retrieves location-based data, and AR-

based assistance. All things considered, the combination of semantic web technology with augmented reality offers a bright future for interior navigation.

Since AR mobile applications require a lot of processing power, their performance may be poor. Furthermore, interior placement is still difficult because of environmental influences, which might lead to incorrect outcomes. Furthermore, because it will need more testing and optimisation, there is no viable implementation in large-scale environments.

3.4.11 Augmented Reality: A Comprehensive Review

The following study by (Shaveta Dargan, 2022) explores augmented reality in further detail and shows how it has changed throughout time in the fields of entertainment, education, healthcare, tourism, and robotics. Typically, sensors in augmented reality collect data from the actual environment, which computers then analyse, interpret, and show on display devices. The study also looks at how overlaying virtual things in real-life settings using augmented reality improves experiences in the actual world.

Additionally, the research study focusses on a number of industries where augmented reality has had a significant influence, including robotics, education, healthcare, entertainment, gaming, and tourism. One of the main factors in enhancing the human experience is the short time needed for interaction between the actual and virtual worlds while using AR. Similar to this, augmented reality (AR) may be used in the military to visualise real-life simulations on the battlefield prior to actual combat. Augmented reality is another tool that may be used to determine a user's position. The user can use AR GPS, compass maps, 3D maps, and other apps to record the location from their GPS and set it in a real environment with changeable size.

Although AR is a game-changing technology, there are a number of risks and restrictions associated with it. For example, adopting AR for gaming has increased the number of fatalities, car collisions, and many other incidents. For example, many young people's mental health has suffered as a result of apps like Pokémon Go. Additionally, latency problems with AR frequently affect real-time engagement. AR application optimisation might be challenging.

3.4.12 Augmenting Reality in Mountain Routes

In the research paper that follows, (Prapas, et al., 2023) explores how augmented reality (AR) technology might improve agritourism by giving consumers an immersive approach to learn about touristic itineraries. In order to give visitors an immersive experience when interacting with routes, the research also focuses on creating a mixed reality application for Microsoft Hololens 2. The purpose of this app is to record tourist interests and to advertise Western Macedonia's mountainous regions.

Hence, the integration of mixed reality and AR can promote mountain routes by significantly increasing tourist engagement by overlaying virtual objects where information regarding the place is displayed in the real world. Through interactive guidance, this app also promotes agritourism in mountainous regions, which are its main attraction. Following that, with the use of geospatial technology, real-time environmental data, and digital route markers can be overlaid. However, because AR and mixed reality work together, it is highly device-dependent, which restricts accessibility for everyone. Furthermore, the accuracy and overall performance may be impacted by the weather. Moreover, considerable computing power is needed for these kinds of applications, which may necessitate substantial battery capacity.

3.4.13 Mont - An Augmented Reality Mountain Peak Recognition Web App in Vanilla Javascript

(Weibel, 2024) study on the following issue, "Mont," a AR-based app for identifying mountain peaks. In the study report that follows, the names of Swiss hilltops and mountain peaks are displayed together with comprehensive details about them, like distance and altitude, using a mobile device's camera feed. Open data created using JavaScript forms the basis of the this research paper. The provided system offers a user interface where users can aim their device's camera at a mountain range. The app overlays the name, heights, and distance automatically. The smooth incorporation of AR improves the user experience. Additionally, the program does not rely on the server side but rather fully dependent on client side.

The "Mont" web application that was delivered was made entirely of vanilla. JavaScript is a groundbreaking AR application; it has a number of drawbacks and unresolved

research needs. Because the software is exclusive to iOS devices, Android users have less access to it. The author should concentrate on developing cross-platform web applications in the future. Furthermore, it emphasizes the efficient use of sensor data. Additionally, the smartphone's built-in magnetic compass module is not very dependable. The author should focus on enhancing smartphone sensors' accuracy in difficult environmental situations.

3.4.14 Incorporating Maps into Flutter: A Study of Mapping SDKs and Their Integration into a Cross-Platform Navigation Application

A quick assessment of different mapping SDKs for incorporation into a Flutter-based navigation application is given in the research paper by (Ushakov, 2024) that is supplied. The study examines a number of maps, including OpenStreetMap, Mapbox, and Google Maps, and verifies their adaptability and interoperability. The study offers a thorough analysis and comparison of several mapping techniques. Additionally, Flutter also presents some integration difficulties.

Furthermore, the study looks at how rendering maps affects an app's overall performance. Furthermore, the study indicates that Google Maps offers significantly greater personalization, a smoother user experience, and slightly better performance. Nevertheless, the study only looks at three kinds of map SDKs. Furthermore, because Google Maps has a high API fee, implementing it might be expensive.

3.4.15 Augmented reality technologies, systems and applications

The research study that is offered aims to give a thorough overview of augmented reality with an emphasis on its technological underpinnings and real-world applications in a variety of sectors. (Julie Carmigniani, 2020) study article below lays the groundwork for many academics and developers to get familiar with the emerging technology known as augmented reality.

The paper begins by providing a quick overview of augmented reality and its classification. It also describes the distinctions between mixed reality and virtual reality and augmented reality. Furthermore, it explains the fundamental elements of augmented reality, including sensors and software layers. In a similar vein, the study notes a number of technological difficulties, including tracking robustness and latency. Additionally,

because AR uses a lot of processing power, the author demonstrates how utilizing it might deplete batteries.

3.5 Summary of Literature Review

The literature analysis above provides a brief explanation of the growing popularity of contemporary technologies such as augmented reality (AR). Numerous research papers have examined the latest developments in augmented reality mobile applications. Furthermore, the concept of overlaying virtual items onto the physical environment is briefly introduced. Since the AR Mount Guide makes use of augmented reality in mobile applications, it can increase the overall experience for tourist as well as regular people in viewing and exploring mountains.

Furthermore, a number of studies highlight the ways in which augmented reality has improved a range of sectors, including tourism, education, healthcare, and entertainment. Additionally, it investigates how augmented reality (AR) might help education by offering real-time information about geographic location, natural heritage, landscapes, and mountainous areas. This improves the quality of instruction and learning. Additionally, by improving the whole visitor experience, the papers offered concentrate on utilizing AR in tourism. Tourists can have a fun and engaging experience when historical tours, interactive route guidance, museums, and art galleries are offered. Furthermore, the user's current location and the distance between them and the mountains can be determined due to the development of location-based services and geospatial technology.

Even though augmented reality has advanced recently, integrating AR with mobile applications is still a very difficult issue. The challenge of integrating augmented reality in unfavorable lighting, climatic circumstances, and many other situations has been clarified by additional research. This may have an impact on how pictures or even 3D models are rendered. Additionally, the literature review supplied outlines all of the restrictions and limits of AR, including the fact that it utilizes a lot of processing power and shortens battery life for AR applications.

3.6 Justification

To guarantee the system's general viability, scalability, and user engagement, the AR Mount Guide project combines a number of contemporary technologies and approaches.

Agile techniques will be applied for this project, and multiple sprints will be established. This will assist in managing intricate modules like mountain recognition, 3D rendering, and many more because of the ongoing and iterative improvement. The agile framework will be applied in a number of stages, including the development, testing, deployment, and prototyping phases.

The prototype design for this application will be made using Figma to create a simple mobile app. Additionally, because of its quick rendering and adaptable user interface components, Flutter will be used to create mobile applications. Additionally, MongoDB and Node.js will be utilized as backend technologies to build a scalable and adaptable database. It also helps in handling geographical data and running sophisticated searches. It is possible to construct a completely scalable application by combining Flutter and Node.js

Since the ARCore package supports motion tracking, environment reliability, and rendering 3D elements, it will be utilized to render AR. In addition, it is available in Flutter. Moreover, the AR Mount Guide will undoubtedly benefit from the lightweight AR models that the ARCore package can create in order to depict mountain terrain. Moreover, lightweight 3D models will be created using Blender. High-quality 3D models can optionally be rendered for this application using Unity. Likewise, this application will use the camera view package to obtain a live camera feed from the user's mobile device. The user's current location, including latitude and longitude, will also be obtained using geolocation. Similar to this, the app will use a gyroscope for accurate orientation and compass sensors to establish the position. Overall, using these technologies, the AR Mount Guide project can be successfully created.

Chapter 4: Primary Research

4.1 Objectives of primary research

Primary research is the method that heavily relies on data collection and is done by the researchers themselves. The main objective of the primary research is to collect and analyse valuable data from the users to know about the AR Mount Guide project (Anon., 2023). The following research aims to gain the following insights:

- To identify the user's familiarity with mountain landscapes.
- To find out which techniques people choose for identifying mountains
- To determine whether users have any prior knowledge or expertise with augmented reality applications.
- To gain and obtain further insightful feedback and recommendations from users to enhance the value of this app
- To learn about the user feasibility in using AR Mount Guide applications.

4.2 Survey Questionnaire

The survey questionnaire technique has been chosen for the AR Mount Guide project. As a result, a Google Form with 15 questions was made and distributed to all users and the intended audience. The purpose of the study is to determine whether the AR Mount Guide app is feasible by examining the survey replies (George, 2023). The list of questions and their purpose are provided below.

S.N	Questions	Objective
1	How often do you go on nature adventure?	To find out the frequency of outdoor activities among users inorder to prioritize features for different age group.
2	Have you ever been to a mountain or hilly area?	To identify how many users are familiar with mountain landscapes
3	Do you find it easy to recognize mountain names?	To analyze whether user struggle with identifying mountains or not

4	When you see a breathtaking mountain, how do you find out more about it?	To understand in what ways users learn and find about the mountains
5	Ever wished for an easier way to instantly know the name of a mountain?	To know user interest in AR Mount Guide core functionality
6	Have you ever used an AR app before?	To know if the users are familiar with AR technology or not
7	What features would make this AR Mount Guide awesome?	To prioritize app based on user preferences in order to increase user experience
8	Would you love to see mountain history and fun facts pop up in AR?	To determine if user would be interested in seeing fun facts pop up in AR Mountain Guide
9	If you could use AR to explore any natural wonder, which one would you choose?	To know about if the user is interested in exploring natural wonder
10	Would you be interested in an indoor Ar Mount Guide experience?	To analyze interests of user who want to explore mountains in a closed room
11	What would make this app a must-have for you?	To understand user point of view on if they are interested in this application.
12	Want to be a beta tester and try it first?	To know if users are ready for beta testing and gather valuable feedback
13	Any fun ideas or crazy suggestions?	To gather any ideas or suggestion in order to improve the application

Table 3 : Survey Questionnaire

4.2.1 Survey Responses

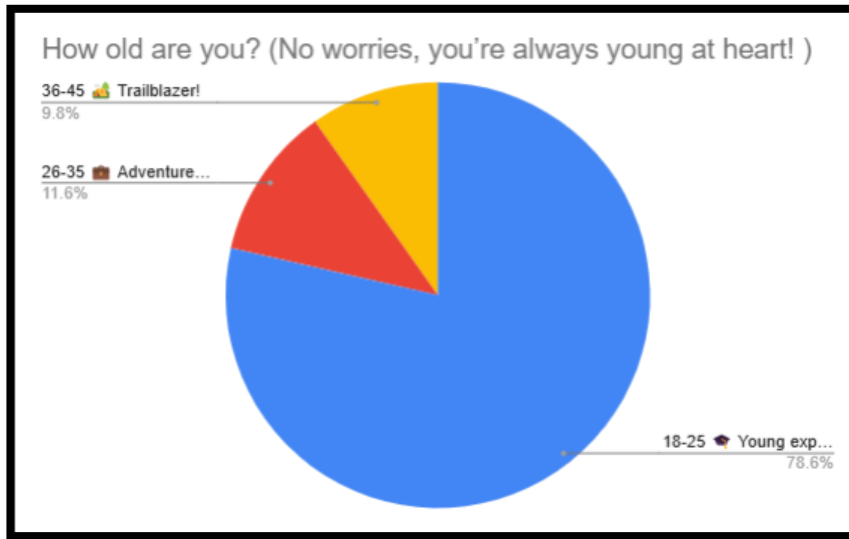


Figure 4: Age group of individual filling this survey

The age distribution of survey respondents is seen in the pie chart below. Approximately 78.6% of the people in this group were explorers between the ages of 18 and 25. In addition, 9.8% of those in the 36–45 age range were trailblazers. Finally, 11.6% of those in the adventure age range of 26 to 35 were in this group

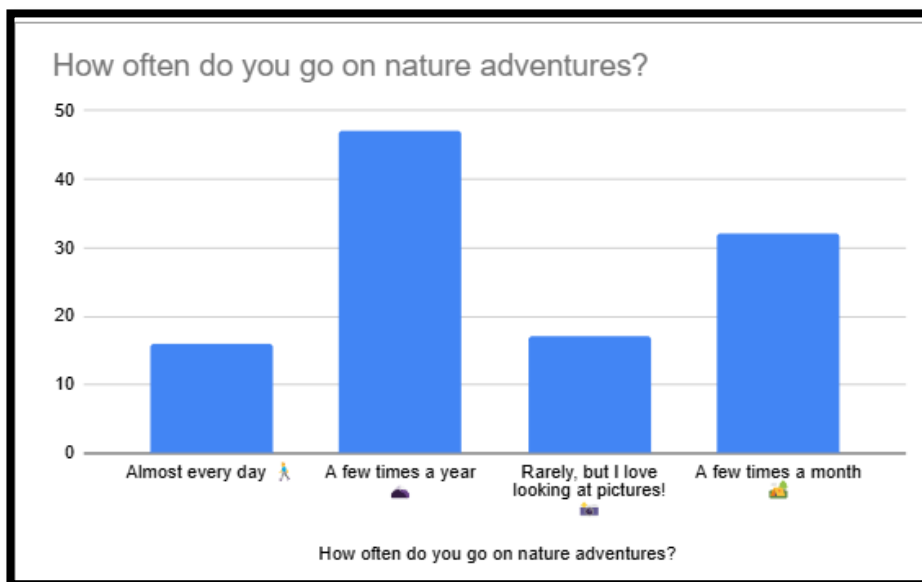
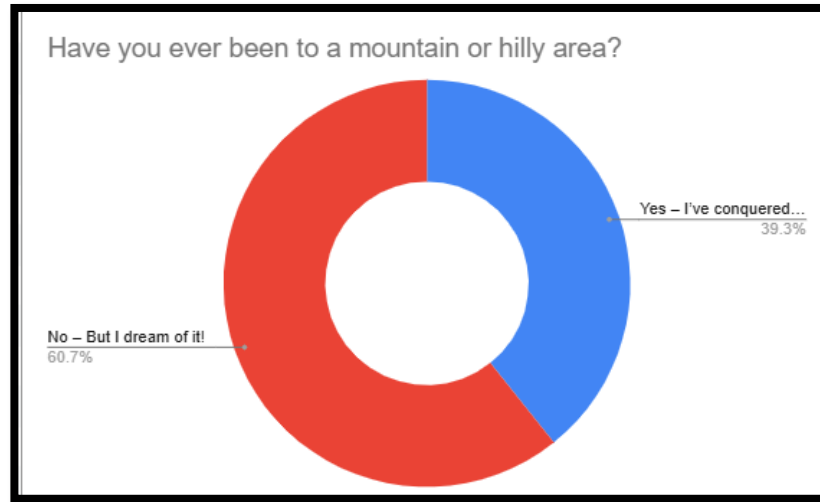


Figure 5: Bar graph - To know on how often individual go to nature

The associated bar graph indicated whether or not a person has ever been to the nature. With around 40 persons, there are more people who have visited nature a few times a year altogether. Additionally, somewhat more than thirty people go to the nature a few times every month. On the other hand, ten to fifteen people go to the nature virtually every day. However, 15–16 individuals are interested in gazing at the photo but hardly ever visit



nature.

Figure 6: Doughnut chart - To know whether user has ever been to mountain or hilly area

The number of persons who have visited the hilly or mountainous regions is depicted in the doughnut chart. About 39.3% of persons in this area have been to a mountain or mountainous location. Nonetheless, about 60.7% of people have only dreamed about visiting mountainous regions.

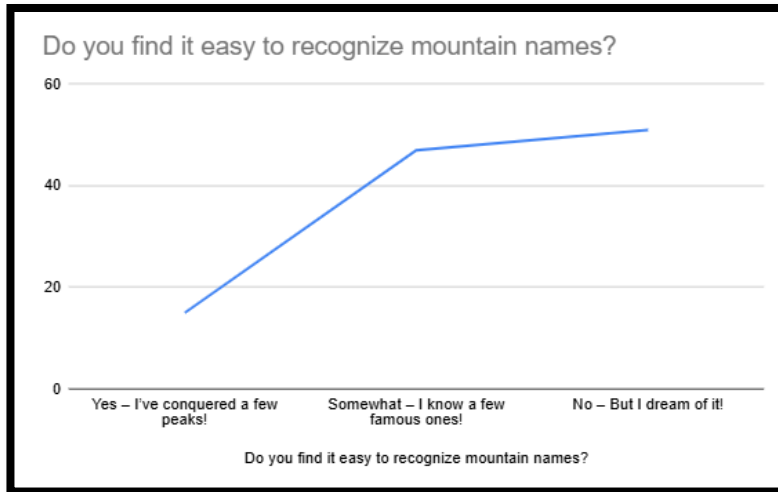


Figure 7: Line Chart - To know if user find it easy to recognize mountains

The line chart illustrates how easy it is for individuals to identify mountain names. Here, fifty people struggle to recognize the mountain names. About 40 participants also had no issue recognizing mountains.



Figure 8: 3D pie chart - To know how individual research on mountain after seeing them

The pie chart that is supplied illustrates the proportion of people who discover more about mountains after witnessing a stunning mountain. When they first see the mountain, about 41.6% of them Google it. Furthermore, 34.5% of people only find the view to be beautiful. Furthermore, about 17.7% of people ask their acquaintances about mountains. About 6.2% of people, however, said they would rather read books.

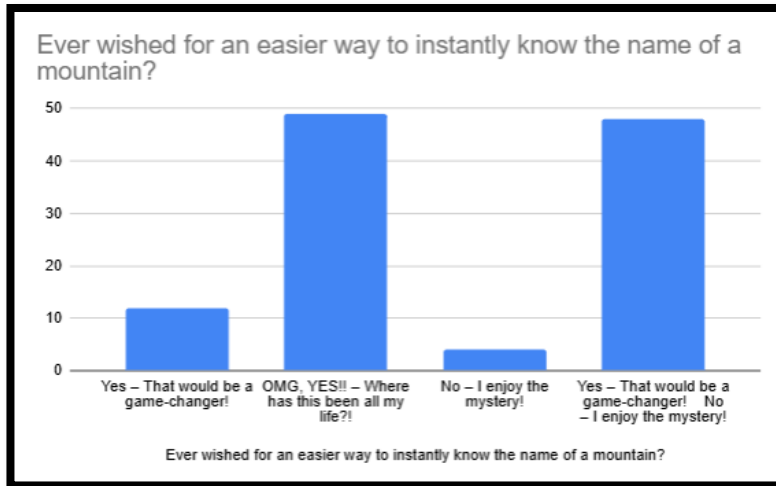


Figure 9; Bar graph - To examine if user would prefer easier medium to know about mountain name

The overall number of people interested in a simpler way to learn about mountains is depicted in the bar graph that is supplied. About fifty people are thrilled about the new, simpler way to learn more about the mountain. Similarly, roughly 45 people believe that finding a simpler way to learn about mountains would be revolutionary. In the same way, ten to fifteen people would appreciate challenging ways to learn more about the mountains.

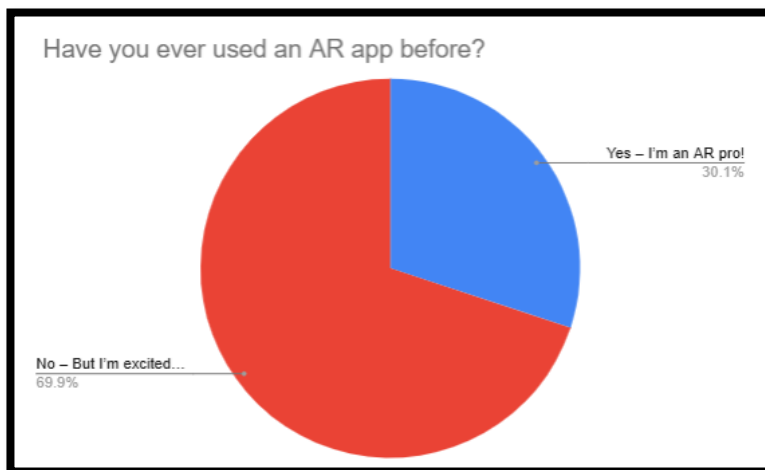


Figure 10: Pie Chart - To know if the uses have used an AR app before or not

The pie chart mentioned above shows whether or not a person has previously utilized an augmented reality app. About 69.9% of people in this survey have never used an

augmented reality app before, but they are eager to do so. Additionally, 30.1% of people had already used an augmented reality app.

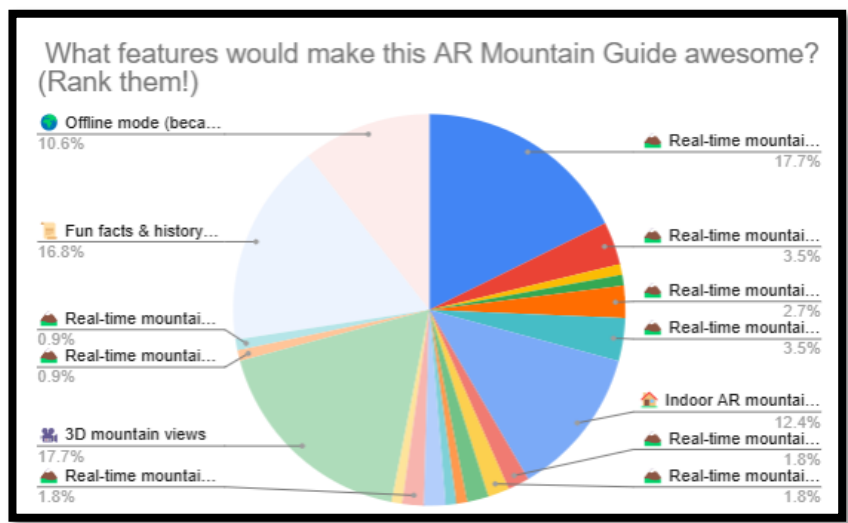


Figure 11: Pie Chart - To identify the features that would entice individual to use this app

The features that people would want to see in the AR Mount Guide app are depicted in the pie chart below. Real-time mountain recognition is the most wanted feature here, with 17.7% of respondents expressing interest. Furthermore, 16.8% of people want to learn interesting historical information. Additionally, 12.4% of people are considering indoor AR Mount Guide simulation. Additionally, about 10% of people are drawn to offline features.

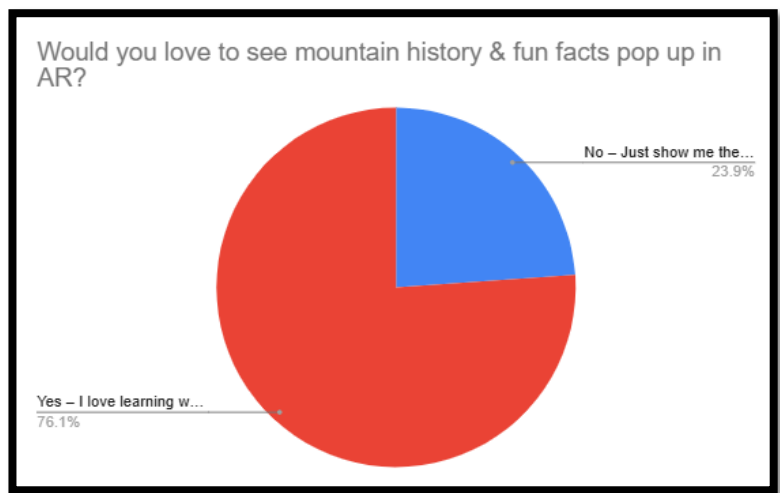


Figure 12: Pie Chart - To learn about user preference in viewing fun facts pop up in AR

The number of people who want to view entertaining popups about the mountain is shown in the pie chart that is provided. Approximately 76.1% of people would want to see fun facts, whereas 23% would rather not.

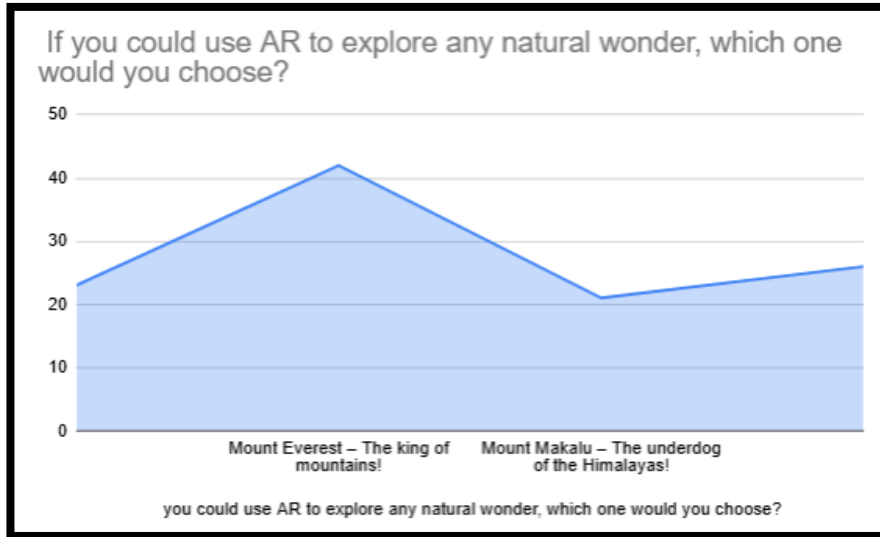


Figure 13: Line Chart - To know on which natural wonder would individual want to explore using AR

Mount Everest and Mount Makalu are the most popular destinations for people to use AR to explore, according to the presented line chart.

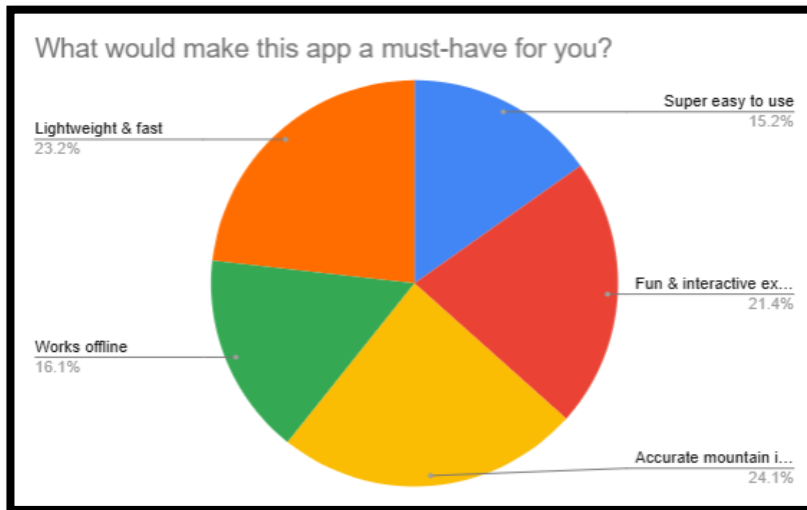


Figure 14: Pie Chart - To examine user preference for this app

The AR Mount Guide application's features that users like best are shown in the pie chart that is provided. In this case, approximately 24.1% and 23.2% of respondents,

respectively, would favor precise mountain identification, lightweight design, and quick application. Additionally, almost 21% of users choose an enjoyable and engaging experience, while 16.1% and 15.2%, respectively, prefer an offline and extremely user-friendly software.

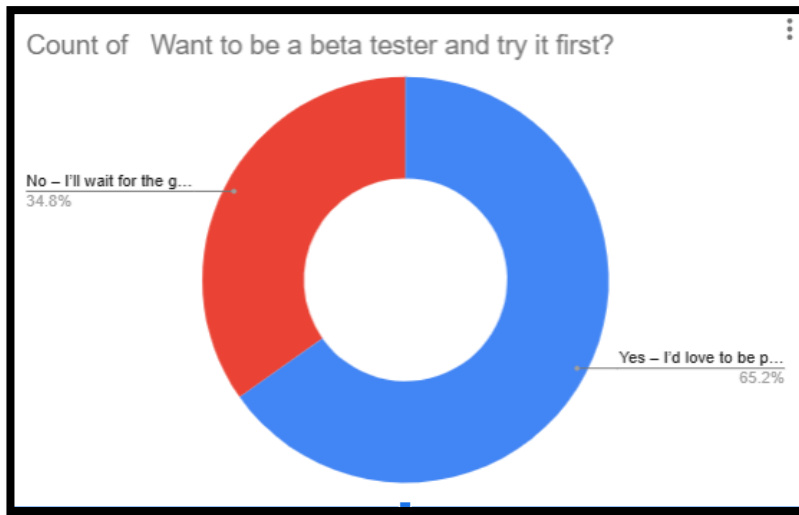


Figure 15: Doughnut Chart - Interested candidate for beta testing the application

The total number of users interested in testing the application is shown in the doughnut chart above. Of those surveyed, around 65.2% would like to try the application, while 34.8% would prefer to wait for its official release.

4.3 Data Collection and Analysis

This chapter's section provides a brief overview of several techniques, including primary and secondary sources, to strengthen and commercialize the AR Mount Guide project (Jain, 2025). A Google form comprising roughly 16 questions was used for primary research in order to gather the data required for the AR Mount Guide project. This survey's primary goal is to collect user interest in AR, issues with AR, and features that users believe are necessary to make the software workable. More clarification on user requirements and how this program can be practical for Individuals was obtained by conducting and collecting overall data from survey responses. In addition, a thorough literature research was carried out on the current technologies, their approaches, important discoveries, and constraints (Journal, 2024). Additionally, it covers current augmented

reality trends and technologies as well as how AR might help the tourism and education industries.

4.3.1 Data Analysis

Quantitative Data Analysis

Concrete data that may be statistically analyzed is provided by the segment that focuses on measuring different aspects of system performance and user interaction. Key measures including accuracy rate, response speed, user engagement metrics, the effectiveness of geographical queries, usability testing results, and scalability are a few examples (Kenton, 2023). While accuracy rate is concerned with correctly identifying mountains, response time is concerned with responding quickly within few seconds. The number of scans per user and the average session length are also key components of user engagement metrics (Taylor, 2022). In order to make sure that the application can support increasing about of users without experiencing a decline in overall performance, scalability is also assessed. In general, these aid in identifying the system's performance strengths and weaknesses.

Qualitative Data Analysis

The survey's results were carefully reviewed and evaluated to learn more about user preferences, technological familiarity, behavioral insights, and perception in general (Dye, 2025). Insightful information about what consumers desire from this project and how to use this app to make it more engaging was obtained by using open-ended responses. Furthermore, although the majority of users have never used an augmented reality (AR) app, they do express interest in embracing new technologies like AR. The majority of users are interested in the main features of the AR Mount Guide app and would like to be beta testers for this project, according to the qualitative comments, which offered profound insights overall (Stewart, 2023).

Chapter 5: Artefact Planning

5.1 Requirement Analysis

5.1.1 Functional Requirements

S.N	Requirements	Priority
1	User must be able register , login and update their profile	MUST
2	User must be able to view list of mountains with relevant information	MUST
3	System must ask user to add their current location via map or geolocation	MUST
4	System must display AR overlay or markers of mountain and its terrain	MUST
5	User must be able to scan their environment	MUST
6	User should be able to use normal mountain overlay for viewing mountains	SHOULD
7	User should be able to view the 3d model of the mountain detected along with its description	SHOULD
8	System should show distance between user and mountain	SHOULD

Table 4: functional requirements

5.1.2 Non Functional Requirements

S.N	Requirements	Priority
1	System must ask user permission for using user current location	MUST
2	System should handle the various weather conditions and constraints	SHOULD
3	System must ask user permission for using camera, GPS and storage of their mobile devices	MUST
4	System could be compatible with both ios and android	SHOULD

5	System should render 3D model and AR smoothly	SHOULD
6	The system should be lightweight and power efficient.	SHOULD
7	The system must be user friendly to use	MUST

Table 5: Non-functional requirements

5.1.3 Usability Requirements

S.N	Requirements	Priority
1	User interface of mobile application should be simple, minimalistic and responsive	SHOULD
2	System should support both light mode and dark mode	SHOULD
3	System should use clear font for better readability	SHOULD
4	System could provide clear interactive tutorial for using this application	COULD
5	System should allow user to provide feedbacks or ask help for any trouble shoot	SHOULD

Table 6: Usability Requirements

5.2 Software Requirements

Operating system

1. Android 7 or more

Programming Language

1. Dart (for flutter)
2. Javascript(for node)

Development Tools

1. Android Studio (for flutter)
2. Android SDK
3. Vs Code (for node)

Framework, Libraries, tools and database

1. Flutter (for creating cross platform mobile application)
2. Express (for backend server)
3. ARKit /ARCore (AR library in flutter)
4. Blender (for 3d model)
5. Geolocation and location services (for real time user location)
6. Cloudinary (for cloud storage)
7. MongoDB (for database)

5.3 Hardware Requirements

5.3.1 Development Requirements

- Minimum of intel core i5 / AMD ryzen 5
- Minimum RAM of 8gb or higher
- Windows 10/11

5.3.2 Deployment Requirements

- Android phone with 4GB RAM or higher
- Minimum of atleast 150-200MB storage for app
- Minimum of 12 megapixel of higher rear camera
- Mobile phone with support of Augmented Reality Components
- Minimum of 3000mAh battery health
- Network support of 4G and 5G

5.4 System Design

5.4.1 Use Case Diagram

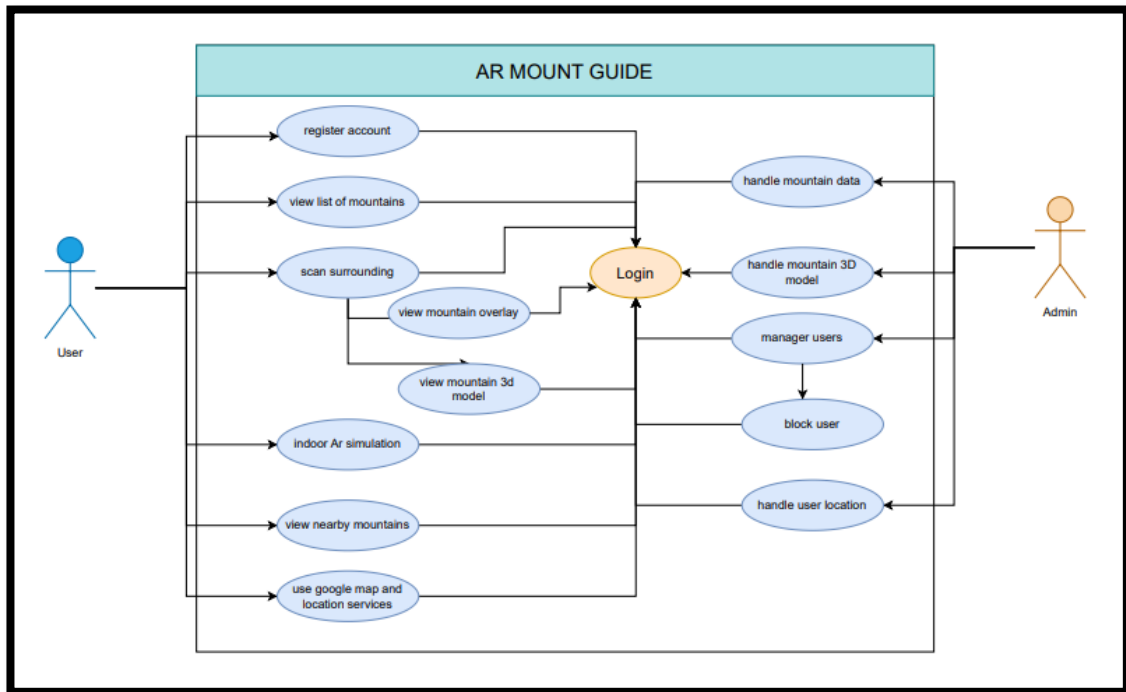


Figure 16: Use case diagram for AR Mount Guide

5.4.2 Database Schema

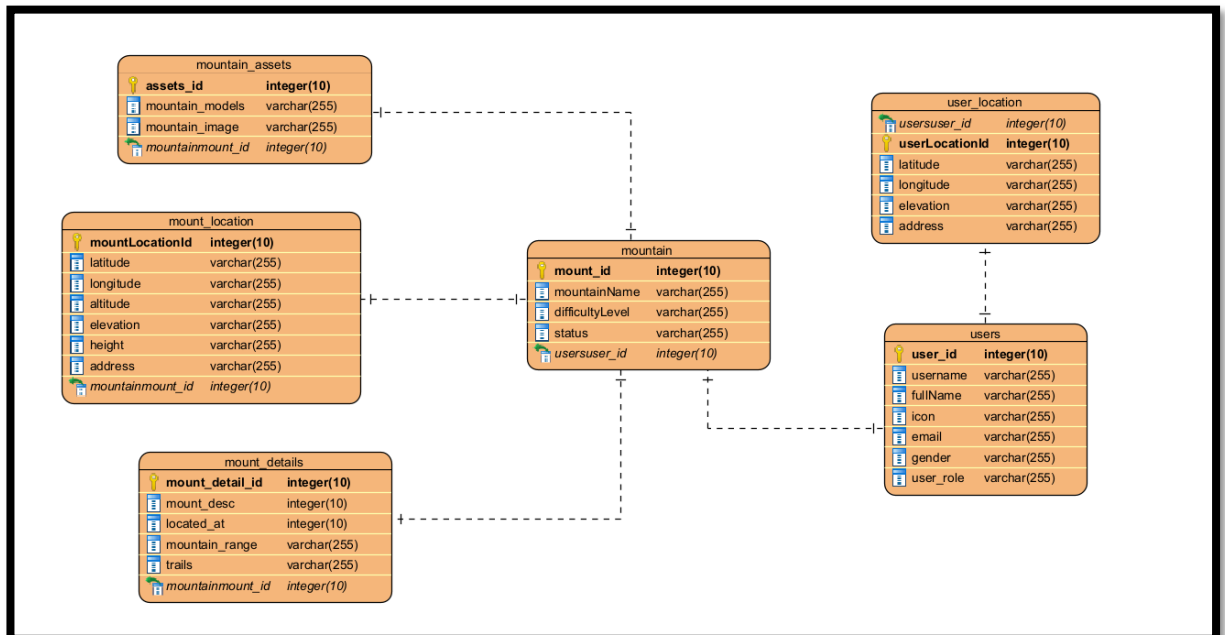


Figure 17: Entity Relationship Diagram

5.4.3 Activity Diagram

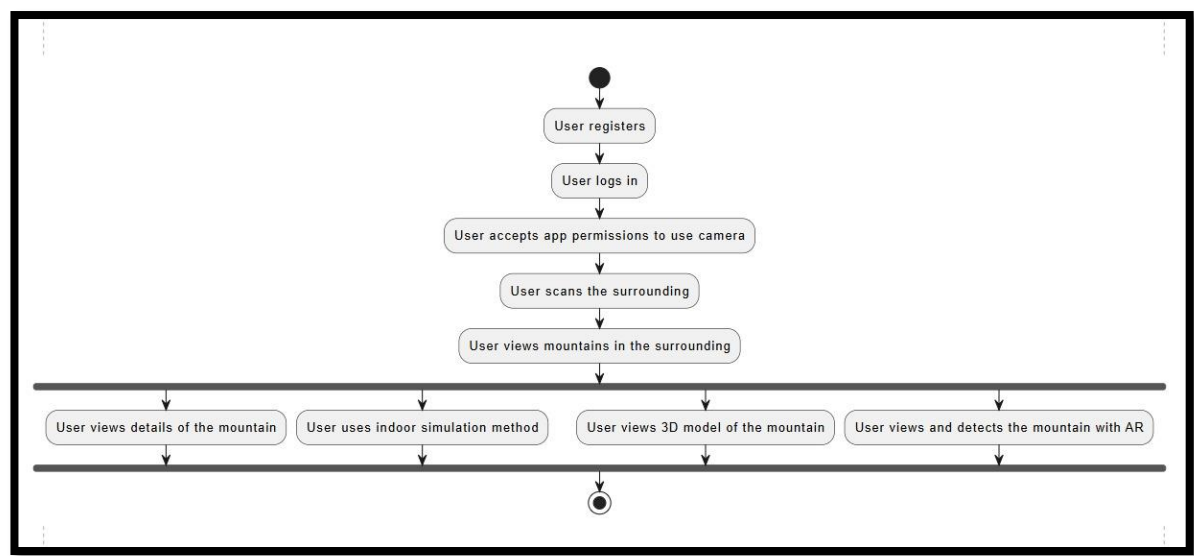


Figure 18: Activity Diagram for AR Mount Guide

Chapter 6: Testing and Evaluation Plan

6.1 Testing Strategy / plan

The thorough plan to guarantee the project's overall success is the testing strategy. In a similar spirit, AR Mount Guide will also go through a number of testing techniques to ensure that all of its features operate effectively (s.Pressman, 2024). The many testing methods suitable for the AR Mount guide project are listed below.

6.1.1 Unit Testing

During unit testing, each code block and component is separated to ensure that the particular function runs error-free (Bakharev, 2024). The test cases for each module will be divided here, including those for the authentication, AR rendering detection, and many more modules. By doing this, unit testing will maintain a good code architecture and streamline debugging, and generally enhance code standards.

6.1.2 Integrated Testing

Integrated testing is done immediately following unit testing. The main objective of integrated testing is to ensure that all the individual components interact correctly, maintaining good workflow (Yasar, 2022). This phase of testing will mainly focus on API integration with the frontend and see whether the system works efficiently, maintaining a good base without showing any sign of error.

Test Cases

Test Case no	Test Case Description	Input	Desired Output
01	Register with valid credentials	Enter valid first name, email , username and password	Account Created successfull !!
02	Register with existing email	Enter email which is already created	Email already exists!!
03	Login with invalid username and password	Enter incorrect values for username and password	Invalid Credentials!
04	Login with valid username and password	Enter valid username and password	Login successfull!
05	Login with empty fields	Enter no values in username and password	Enter your username and email

06	Fetch mountain data from backend	Launch app and scan surrounding and fetch mountain	Mountain identification based on user preferred location
07	Retrieve User Coordinates	Ask user permission to get access to user location	Get accurate user coordinates
08	Tap Marker to view mountain details	Tap on detected mountain	Mountain 3D model appears with description

Table 7: Integrated Testing (test cases)

6.1.3 System Testing

System testing is necessary to determine whether or not all of the integrated components operate quickly and effectively (Das, 2024). To ensure that the system is dependable in all edge case scenarios, this kind of testing is required. Prior to user acceptability testing, this testing will guarantee that both the functional and non-functional criteria of the AR Mount Guide application are met effectively.

Test Cases

Test Case no	Test Case Description	Input	Desired Output
01	Scan Mountain	Point mobile camera in any direction	Accurate identification of mountain with mountain name
02	Disagree for location usage	Disagree when asked permission for location	Show message : Location is required before you move ahead
03	Smooth 3D rendering	Use indoor simulation or click on identified mountain	No lags and 3d model is rendered effectively
04	Accurate fetching of mountain data	Fetch mountain data when scanned on surrounding	Data is accurately fetched from the database

Table 8: System Testing (test cases)

6.1.4 User Acceptance Testing

This is the last testing plan for the AR Mount Guide project, in which actual users will test the program in authentic settings (Alimi, 2020). The purpose of the subsequent test is to gather user input and ensure that the product is dependable prior to deployment. The application is prepared for release if it receives positive feedback.

Test Case no	Test Case Description	Description	Desired Output
01	Users scan with mobile camera	Opens app and scan the surrounding or mountain	Accurate mountain name is displayed
02	Users view mountain information	Clicks on identified mountain	View 3d model of mountain along with mountain information
03	Users utilizes indoor simulation	Users sits in closed room and scan surrounding	Accurate mountain name is displayed

Table 9: User Acceptance Testing (Test cases)

6.2 Evaluation Strategy / plan

The main objective of the evaluation strategy is to evaluate the overall usability and effectiveness of AR Mount Guide features (Wright, 2023). The DECIDE framework is utilized for evaluation strategy in order to determine, evaluate, and analyze the overall goal and how to present it (Student, 2022).

Determine the goals

- Analyze overall usability of Ar Mount Guide.
- Ensure mountain idention works as expected.
- Ensure indoor simulation to work efficiently.
- Check whether users are satisfied with this application.

Explore the questions

- Do users understand the AR technology?
- Does the app meet user expectations?
- How effective is the Ar Mountain Detection strategy?
- Are AR labels of mountains accurate?
- Is the provided information about the mountain correct and accurate?

Choose the evaluation methods

- User Evaluation Testing: Users will be provided to test this application to gather the necessary feedback.
- Testing Strategy: Several testing strategies, such as system testing, unit testing, and integrated testing, will be conducted to ensure the system's overall performance.
- Survey and Questionnaire: A Google form will be created and sent to people in order to collect data and analyze if they are satisfied with the given product or not.

Identify Practical Issues

- Ensure safe areas of testing AR Mount Guide with proper lighting conditions.
- Detailed guidelines and a tutorial will be provided for users on how to use the following application.
- Testing of the application will be done in both in-person and hybrid.

Decide how to deal with the ethical issues

- Maintain data privacy of users.
- Take user permission before accessing their camera.
- Also get user location only if they agree to share their location.

Evaluation

- All the feedback collected during the evaluation will be effectively reviewed and analyzed. Data collection will be done based on that, and all the suggestions for improvement will be noted down in order to make the app more user-friendly.

Chapter 7: Development Process

The augmented reality-based Ar Mount Guide application was developed using a methodical, multi-phase strategy that comprises geolocation processing, device sensors, AR environment setup, mountain data, distance computation, and real-time augmented overlay presentation. Flutter is used in this project to create a mobile application. Real-time mountain identification, a smooth, user-friendly mountain overlay, and an improved method of delivering mountain overlay are all part of the development process. The whole development process for development stages such as ar setup, distance calculation and ar mountain overlay are explained below.

7.1 Methodology

The given project makes use of the fundamental idea of the agile approach, which facilitated the iterative development process. Augmented reality, sensor integration, real-time computations, and AR overlay are all part of the project. Agile approach greatly aids in dealing with it by enabling prompt adaption to any technological obstacles and by providing ongoing supervisory input.

7.2 Tech stack utilized

Hardware: Android phone with google arcore installed

Software

- Flutter for mobile application
- Node.js and mongodb as database

7.3 Development Planning and Data Collection

Effective planning was therefore maintained for a smooth development process by outlining all the features and requirements, putting them into practice, and working on project development. A few things were changed, but I was still able to keep at a fair pace because to the effective prioritization. To complete the project, each task was successfully divided and addressed one at a time. Additionally, I used Wikipedia to gather all of the

information on mountains for the data collection, but I only used data from ten of them. So these data are accurate according to Wikipedia.

```
static const List<Map<String, dynamic>> _mountains = [  
  {"name": "Mount Everest", "lat": 27.9881, "lon": 86.9250, "ele": 8848.86},  
  {"name": "Lhotse", "lat": 27.9617, "lon": 86.9330, "ele": 8516},  
  {"name": "Makalu", "lat": 27.8897, "lon": 87.0888, "ele": 8485},  
  {"name": "Cho Oyu", "lat": 28.0942, "lon": 86.6608, "ele": 8188},  
  {"name": "Manaslu", "lat": 28.5497, "lon": 84.5597, "ele": 8163},  
  {"name": "Annapurna I", "lat": 28.5958, "lon": 83.8203, "ele": 8091},  
  {"name": "Champa Devi", "lat": 27.6535, "lon": 85.2655, "ele": 2241},  
  {"name": "Langtang Lirung", "lat": 28.2117, "lon": 85.5125, "ele": 7227},  
  {"name": "Gaurishankar", "lat": 27.9597, "lon": 86.3686, "ele": 7134},  
  {"name": "Dhaulagiri", "lat": 28.6967, "lon": 83.4878, "ele": 8167},  
];
```

Figure 19: Data collected for mountain

7.4 Managing android permission and project initialisation

Therefore, maintaining a number of permissions was essential for developing augmented reality mobile applications. For example, maintaining permissions was necessary for utilizing the internet, camera access, and AR camera. Further the app also ask permission regarding the location, camera and microphone with the help of permission handler. The Android Manifest file contains these permissions. In a similar vein, Flutter was used to build up the project, and all required packages were installed for the application to function properly.

```
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />  
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />  
<uses-permission android:name="android.permission.ACCESS_BACKGROUND_LOCATION" />  
<uses-permission android:name="android.permission.CAMERA" />  
<uses-permission android:name="android.permission.RECORD_AUDIO" />  
<uses-permission android:name="android.permission.INTERNET" />
```

Figure 20:Permission Mentioned for application

```
cupertino_icons: ^1.0.8
camera: ^0.11.0+2
sensors_plus: ^6.1.1
geocoding: ^4.0.0
logger: ^2.6.2
geolocator: ^13.0.2
flutter_compass: ^0.8.1
model_viewer_plus: ^1.9.3
http: ^1.4.0
arcore_flutter_plugin:
  git:
    url: https://github.com/mystickminal14/arcore_flutter_plugin.git
    ref: master # or your branch name
permission_handler: ^12.0.1
vector_math: ^2.1.4
```

Figure 21: Essential packages for flutter application

7.5 ArCore Integration

So for the augmented reality application, basically, there are two types of AR packages: they are ARCore and ARKit. ARKit is not utilized in this project since it was for iPhone, and due to hardware limitations, it was difficult to continue with ARKit. Nonetheless, ARCore was utilized for this project, and the initial setup was full of bugs and errors. Since Google did not maintain ARCore plugin for 2 years and with limited documentation and community, it was very difficult to continue ARCore with Flutter. However, there was only one solution to this, which was mentioned in GitHub, where the best advice was to custom integrate the GitHub repository directly into the package. This not only solved the AR-related issue but also helped in the overall application.

```
arcore_flutter_plugin:
  git:
    url: https://github.com/mystickminal14/arcore_flutter_plugin.git
    ref: master # or your branch name
```

Figure 22: Custom git integration of arcore flutter plugin

7.6 Ar Rendering and mountain overlay

Therefore, real-world data from sensors is necessary for AR to function properly. It extracts the user's current position, for example. Additionally, real-time heading was achieved by subscribing to compass events using Flutter Compass. For efficient AR display, an AR controller was also initialized. In a similar vein, a model of a mountain with the features required for a mountain overlay was made.

Therefore, the app determines whether the user has moved every ~900 ms in `onArCoreViewCreated`. If so, it uses geolocation formulas to determine the bearing, or direction to the mountain, as well as the distances for each mountain. Therefore, spherical trigonometry was used for bearing and the haversine formula for calculating distance.

To calculate the true distance between the user's current location and each mountain's geographic coordinates, the Ar Mount guide application mainly uses distance computation. The calculation ensures that mountains that are closer are seen in augmented reality, while those that are farther away are positioned deeper in the scene. The Haversine formula, which determines the distance between two points on the planet's surface (user location and mountain position), approximates the great circle distance, which is the shortest trip across Earth's curved surface. The Haversine formula is suitable for geographic coordinates, assuming a spherical Earth with a radius of around 6371 kilometers.

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right)$$

$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R \cdot c$$

Figure 23:Haversine formula

```
void _calculateMountainData() {
    if (_currentPosition == null) return;

    for (final mountain in _mountains) {
        final bearing = _bearingDeg(
            _currentPosition!.latitude,
            _currentPosition!.longitude,
            mountain.latitude,
            mountain.longitude,
        );
        final distance = _haversineKm(
            _currentPosition!.latitude,
            _currentPosition!.longitude,
            mountain.latitude,
            mountain.longitude,
        );

        _mountainBearings[mountain.name] = bearing;
        _mountainDistances[mountain.name] = distance;
    }
}

static double _bearingDeg(double lat1, double lon1, double lat2, double lon2) {
    final dLon = _toRad(lon2 - lon1);
    lat1 = _toRad(lat1);
    lat2 = _toRad(lat2);
    final y = sin(dLon) * cos(lat2);
    final x = cos(lat1) * sin(lat2) - sin(lat1) * cos(lat2) * cos(dLon);
    final brng = atan2(y, x) * 180.0 / pi;
    return (brng + 360) % 360;
}

static double _haversineKm(double lat1, double lon1, double lat2, double lon2) {
    const r = 6371.0; // km
    final dLat = _toRad(lat2 - lat1);
    final dLon = _toRad(lon2 - lon1);
    final a = sin(dLat / 2) * sin(dLat / 2) +
        cos(_toRad(lat1)) * cos(_toRad(lat2)) * sin(dLon / 2) * sin(dLon / 2);
    final c = 2 * atan2(sqrt(a), sqrt(1 - a));
    return r * c;
}
```

Figure 24: Distance Calculation formula

A variety of methods are also used to deliver AR experiences, such as bearing computation for directional alignment, visibility filtering based on device field of view, and AR object placement. Bearing calculation, which determines the compass heading from the user's current position to each mountain, allows AR items to be accurately positioned with regard to real-world orientation. The bearing is calculated using the forward azimuth formula, which establishes the angle between the user position and provides the beginning direction along the great circle path.

$$\begin{aligned}y &= \sin(\Delta\lambda) \cdot \cos(\phi_2) \\x &= \cos(\phi_1) \cdot \sin(\phi_2) - \sin(\phi_1) \cdot \cos(\phi_2) \cdot \cos(\Delta\lambda) \\ \theta &= \text{atan2}(y, x) \\ \text{bearing} &= (\theta \cdot 180/\pi + 360) \bmod 360\end{aligned}$$

Figure 25: Forward Azimuth Formula

A variety of methods are also used to deliver AR experiences, such as the positioning of AR objects, visibility filtering based on the field of view of the device, and bearing computation for directional alignment. Bearing calculation, which determines the compass heading from the user's current position to each mountain, allows AR items to be precisely positioned with respect to real-world orientation. The bearing is calculated using the forward azimuth formula, which establishes the angle between the user's position and the beginning direction along the great circle path. Below is the code snippet:

```

List<String> _getVisibleMountains() {
  final visible = <String>[];
  if (_currentPosition == null || _currentAzimuthDeg == null) return visible;
  for (final mountain in _mountains) {
    final bearing = _mountainBearings[mountain.name];
    if (bearing == null) continue;

    final delta = _normalizeDeg(bearing - _currentAzimuthDeg!);
    if (delta.abs() <= _kFovDeg / 2) {
      visible.add(mountain.name);
    }
  }
  visible.sort((a, b) {
    final bearingA = _mountainBearings[a];
    final bearingB = _mountainBearings[b];
    final deltaA = _normalizeDeg(bearingA - _currentAzimuthDeg!);
    final deltaB = _normalizeDeg(bearingB - _currentAzimuthDeg!);
    return deltaA.compareTo(deltaB);
  });
  return visible;
}

```

Figure 26: get visible mountains

```

vm.Vector3 _calculateARPosition(double bearing, double realWorldDistanceKm) {
  final bearingRad = (bearing - _currentAzimuthDeg!) * pi / 180.0;

  // Map real-world distance to AR distance (logarithmic scale works well)
  final maxRealWorldKm = 500.0; // Adjust based on your mountain distances
  final normalizedDistance = realWorldDistanceKm / maxRealWorldKm;
  final arDistance = _kMinDistance + normalizedDistance * (_kMaxDistance - _kMinDistance);

  // Calculate X and Z coordinates (Y is up in ARCore)
  final x = sin(bearingRad) * arDistance;
  final z = -cos(bearingRad) * arDistance; // Negative Z goes into the screen

  return vm.Vector3(x, 0, z);
}

```

Figure 27: Bearing Calculation

A 3D form with labels that are dynamic PNG textures with text and cached for efficiency is shown on the screen for each visible mountain sphere. Text is rendered onto PNG image using Flutter Canvas and TextPainter creating a texture for an AR plane.

7.7 Frontend Display

As previously said, Flutter was used to complete the frontend. I made two distinct pages for the AR scan and the mountain overlay, which is a regular overlay. AR is used in AR overlay, however it is not used in normal overlay. The photos of the identified mountain and its overlay are shown below.

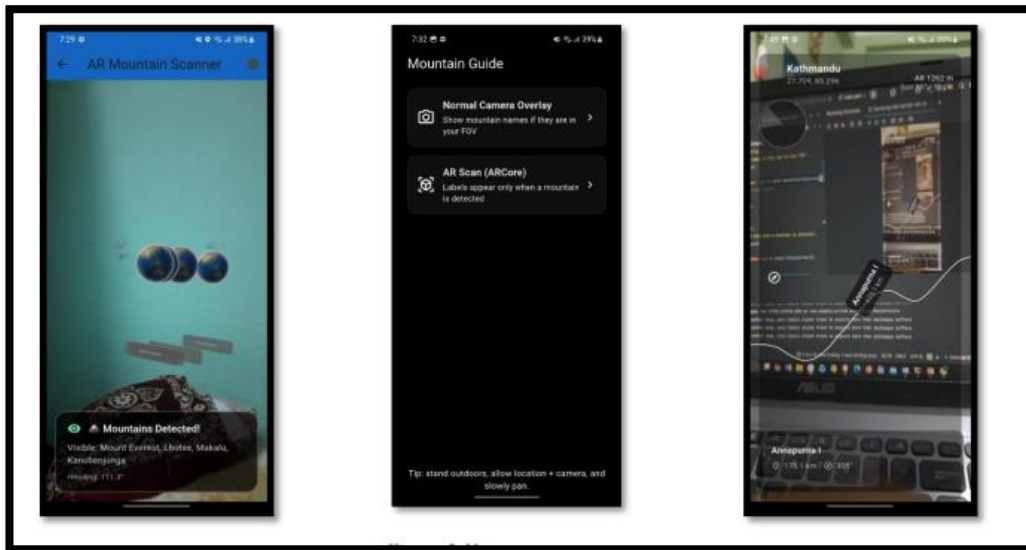


Figure 28: Ar Mountain Overlay

7.8 Backend Integration and interoperability

Thus, I used MongoDB as the database and Node.js and Express.js for the backend. Because the frontend is Flutter and the backend is JavaScript, it makes use of the idea of interoperability because Flutter uses the JavaScript-created API. I have thus used a login and registration system for the backend and frontend connection, allowing users to both register and log in. In a similar vein, it uses a JWT token to maintain correct authentication.

Chapter 8: Testing

The Ar mount guide application underwent a number of testing to ensure that the system functions successfully and efficiently. Unit testing, integrated testing, integration testing, and system testing are only a few of the testing methods used in the project.

8.1 Unit Testing

8.1.1 Unit Test Case Scenario for register

Test Cases No	Test Description	Expected Outcome	Actual Output	Refer to
1	When user sends request with missing fields, it should return 400 status	All fields are required with 400 status	Passed ("All fields are required" ,Status: 409)	Figure 32: Missing fields during register
2	When user sends request with missing username, it should return 400 status	All fields are required with 400 status	Passed ("All fields are required" ,Status: 409)	Figure 33: Missing username during register
3	When user sends request with missing email, it should return 400 status	All fields are required with 400 status	Passed ("All fields are required" ,Status: 409)	Figure 34: Missing email during register
4	When user sends request with missing password, it should return with status 400	All fields are required with 400 status	Passed ("All fields are required" ,Status: 409)	Figure 35: Missing password during register
5	When user enters password less than 8 characters, it should return password is less than 8 character with 400	All fields are required with 400 status	Passed ("All fields are required" ,Status: 409)	Figure 36: Password length
6	Returns status code with 409 if user already exists	User already exists with 409 status	Passed ("User already exists", Status: 409)	Figure 37: User Already exists
7	Successful resgistration	User Register successfully	Passed ("User registered successfully)	

		with 201 status code		
--	--	----------------------	--	--

Table 10: Unit testing for register

8.1.2 Unit Test Case Scenario for login

Test Cases No	Test Description	Expected Outcome	Actual Output	Refer to
1	When a user does not enter an email, it should return "email and password required" with a status of 400.	Email and Password are required with 400 status	Passed ("Email and password are required", Status: 400)	Figure 38: missing email during email
2	When a user does not enter a password, it should return "email and password required" with a status of 400.	All fields are required with 400 status	Passed ("All fields are required are required", Status: 400)	Figure 39: missing password during login
3	When user enters invalid credentials for email and password, it should return 401 with invalid credentials	Invalid credentials	Passed ("Invalid Credentials", Status: 401)	Figure 40: invalid credentials
4	When user enters incorrect password	Invalid password	Passed ("Invalid password", Status: 401)	Figure 41: incorrect password
5	Successful login with 200 status	Login Successful with 200 status	Passed ("Login successful", Status: 200)	Figure 42: successful login

Table 11: Unit test case scenario for login

```

PASS src/tests/user.test.js
Auth Controller Tests
  ✓ should return 400 if fields are missing during register (45 ms)
  ✓ should return 400 if username is missing during register (9 ms)
  ✓ should return 400 if email is missing during register (9 ms)
  ✓ should return 400 if password is missing during register (8 ms)
  ✓ should return 400 if password is less than 8 characters during register (6 ms)
  ✓ should return 409 if user already exists (16 ms)
  ✓ should return 400 if email is missing during login (9 ms)
  ✓ should return 400 if password is missing during login (6 ms)
  ✓ should return 401 for invalid credentials (user not found) (34 ms)
  ✓ should return 401 for incorrect password (14 ms)
  ✓ should login successfully (15 ms)

Test Suites: 1 passed, 1 total
Tests: 11 passed, 11 total
Snapshots: 0 total
Time: 1.588 s, estimated 2 s

```

Figure 29: Test Case Image for login and registration

8.1.3 Unit Test Case Scenario for Ar Mountain Overlay

Test Cases No	Test Description	Expected Outcome	Actual Output	Refer to
1	Device support for ArCore	App detects ArCore support and continue with application	Passed (ArCore initialized successfully)	Figure 43: normalise deg test
2	Calculate bearing between two coordinates	Bearing calculation of Kathmandu – Everest and output should be accurate	Passed Returned accurate answer	Figure 44: bearing test
3	Calculate haversine distance formula	Calculation of Kathmandu – Everest should be almost 65 deg to 75 deg	Passed Returned 65-75deg	Figure 45: haversine test
4	Angular distance between 350 deg and 0 deg	It should be equal to 10	Passed Returned 10	Figure 46: angular diff deg test
5	Calculate Ar Position	Calculating AR x/z coordinate and result should be close to -15	Passed Z returned close to -1.5	Figure 47: calculate ar position test

6	Mountain color	Color for same name should be identical	Passed Color for same name is identical	Figure 48: get mountain color test
7	Label Texture Generation	size of png should be greater than 1000 bytes	Passed PNG size is greater than 1000 bytes	Figure 49: make test png
8	Test visible mountains detection when azimuth =0 deg and bearing is 10 deg	Filters list of mountains within field of view	Passed Returned list of detected mountains	Figure 50: get visible mountains
9	Verify calculateMountainData	Mountain bearing should not be null	Passed Returned null	Figure 51: calculate mountain data
10	Check mountains closes to azimuth	Item should appear sorted	Passed	Figure 51: calculate mountain data

Table 12: Unit test case scenario for ar mountain overlay

```

D:\ar>flutter test
'crumb' is not recognized as an internal or external command,
operable program or batch file.
00:08 +0: Geometry Helpers normalizeDeg
✓ normalizeDeg Test Passed
00:08 +1: Geometry Helpers bearingDeg
✓ bearingDeg Test Passed
00:08 +2: Geometry Helpers haversineKm
✓ haversineKm Test Passed
00:09 +3: Geometry Helpers angularDiffDeg
✓ angularDiffDeg Test Passed
00:09 +4: AR Position Calculation calculateARPosition
✓ calculateARPosition Test Passed
00:09 +5: Mountain Color getMountainColor
✓ getMountainColor Test Passed
00:09 +6: Label Texture Generation makeTextPng
✓ makeTextPng Test Passed
00:09 +7: Visible Mountains getVisibleMountains
✓ getVisibleMountains Test Passed
00:09 +8: Mountain Data Calculation calculateMountainData
✓ calculateMountainData Test Passed
00:09 +9: (tearDownAll)
🎉 ALL TEST CASES PASSED SUCCESSFULLY 🎉
00:09 +9: All tests passed!

```

Figure 30: Flutter test for ar mountain overlay

8.2 System Testing

Test Cases No	Test Description	Expected Outcome	Actual Output
1	Ask permission for camera and ar	If allowed permission it should say “ allowed permission	Permission is permitted
1	Clicks on Normal Overlay	Should open normal overlay screen	Normal overlay screen opened successfully
2	Scans surrounding to detect mountain	Should show mountain overlay of detected mountain	Mountain name and label displayed in the screen
3	Clicks on Ar overlay screen	Should show ar screen	Ar screen opened in screen
4	Scans surrounding to detect mountains in ar	Should shows 3d shapes of sphere with mountain label on it	Detects the mountain and display 3d sphere
5	Clicks on detected mountain label and redirects to mountain details page	Should redirect to mountain detail screen with 3d model	Page redirected to mountain detail screen
6	Interaction with 3d model	User can rotate 3d model	3d model can be rotated in 360 deg

Table 13: System testing for app

8.3 Integrated Testing

Test Cases No	Test Description	Expected Outcome	Result
1	User register , logs in and get access to dashboard	User registration succeeds, login is also successful and user gets access to dashboard	Passed
2	User opens normal overlay and also view detected mountain	Normal overlay should open as well as he can and should display detected mountains	Passed
3	Users opens ar overaly and also scans in ar overlay to display ar object	Should scan surrounding and gets ar objects displayed in the surrounding	Passed
4	Clicks on ar object and open 3d model of mountain as well as description	When clicked on ar object, user should be able to view 3d model as as well as detailed description	Passed

Table 14: integrated testing for app

Chapter 9 : Evaluation

The total assessment of the Ar Mount Guide app based on user evaluation, performance, and dependability is shown in this chapter. Evaluation is essential to ensuring that software can benefit all users since it offers pertinent feedback that may be vital prior to deployment.

9.1 User Evaluation

Real users were given access to the app in order to gather feedback for the AR Mount Guide user evaluation. Everybody had their own opinions, with some praising and some criticizing the application. A overview of user remarks and rating(up to 5) according to criterion is shown in the table.

S.N	Criteria	Remarks	Overall Rating
1	UI/UX design of application	- The user interface design follows modern design standards - Design of Ar overlay looks visually appealing - Ar object feels visually heavy - The home screen option are well organized - The detection card overlaps the ar screen - Can improve the ar mountain details page - Home screen looks very simple and with the addition of dark mode and light mode feature, the app can definitely improve	4
2	User experience of application	- The app has easy and smooth navigation - When rendering the 3d model, the app takes around 5 second and is very uncomfortable while waiting for the 3d - Home screen has clear instruction for normal and ar overlay	3.5

		- Can improve the performance of ar objects placement - There is no use of many color and the app sticks with its core theme which is easier to use	
3	Ar detection	- The ar detection of mountains around me is a revolutionary idea - Excellent application for mountain detection - The detection is good but with the addition of more number of mountain, the app can highly benefit - Smooth rendering of Ar object can be an improvement - Good concept but need to improve the ar object placement - The app gets slow with increasing number of mountains detection	4
4	Normal Overlay	- Normal overlay is faster than ar overlay, however the app can benefit if the design of normal overlay is simple - Design of normal overlay feels congested and can definitely improve with simple design - When multiple mountain are detected, the detected 2d label placement get overlapped with each other	4.5
5	Mountain Detail Screen	- In the detailed screen, the 3d model takes longer time to render. - For every detected mountain, the 3d model is same and can definitely improve in future	4

		- Clean design for detailed screen, displaying every details of mountains - Mountain detail screen is best part due to its interactive 3d model - Can add more details in the mountain details	
6	Authentication	- Secure login and register process - Can improve if it is integrated with google authentication - Works fine , no issue with authentication - Can definitely improve the design of login and register page	4.5

Table 15: User evaluation table

9.2 Performance Analysis

Performance Analysis is essential for AR Mount guide since it determines how fast the system works, how many request a system can handle and also to identify bottle necks and maintain robustness in the application. Here performance analysis is necessary for the system to work efficiently. So for the performance analysis artillery is used since it is well suitable for Node js.

```
1  config:
2    target: "http://localhost:3000"
3    phases:
4      - duration: 60
5        arrivalRate: 20
6        name: "Steady load"
7
8    defaults:
9      headers:
10       Content-Type: "application/json"
11
12  scenarios:
13    - name: "User Registration Load"
14      flow:
15        - post:
16          url: "/api/users/register"
17          json:
18            username: "testuser_{{ $randomString(5) }}"
19            email: "test{{ $randomString(5) }}@example.com"
20            password: "Password123"
21
22    - name: "User Login Load"
23      flow:
24        - post:
25          url: "/api/users/login"
26          json:
27            email: "testuser@example.com" # Replace with a test account
28            password: "Password123"
29
30    - name: "Refresh Token Load"
31      flow:
32        - post:
33          url: "/api/users/refresh-token"
34          json:
35            refreshToken: "{{ refreshToken }}" # Optional: set dynamically if needed
36
37    - name: "Get Current User Load"
38      flow:
39        - get:
40          url: "/api/users/current-user"
41          headers:
42            Authorization: "Bearer {{ accessToken }}" # Optional: set dynamically if needed
43
44    - name: "User Logout Load"
45      flow:
46        - post:
47          url: "/api/users/logout"
48          headers:
49            Authorization: "Bearer {{ accessToken }}" # Optional: set dynamically if needed
50
```

Figure 31: Performance Analysis using artillery

The Node.js test that Artillery ran is shown in the code snippet. When this file was run, a JSON file and an HTML file were produced. Thus, the report was produced with the aid of Grok AI. The Node.js API, which simulated 1200 virtual users over around 7 seconds using 1200 HTTP requests, is summarized in the report. The report that is produced is as follows:

System load test summary:

SN	Metric	Description	Value
1	Duration of Test	Total test running time	70 seconds
2	Total requests sent	Adding up all the http requests	1200
3	Successful responses	Number of successful response	125
4	Failed responses	Number of failed responses	965

5	Total timed out requests	Number of requests that time out	110
6	Failure rate	Rate of failure	4.1%
7	Request per second	Request per second	10
8	Peak concurrent connections	Maximum number of concurrent connections	N/A
9	Total Data Transferred	Combine size of response bodies	0.13 mb
10	Average payload	Average size of response bodies	125byte

*Table 16: System load test summary***Latency and Http Code distribution:**

SN	Metric	Value
1	Average Response time	1774.40 ms
2	Fastest response time	1 ms
3	Slowest Response time	9972.00 ms
4	P50	699.ms
5	P90	5711.50 ms
6	P95	7117.0 ms
7	P99	8352 ms

Table 17: Latency

SN	Metric	Count
1	201	125`
2	401	965
3	4xx/5xx	0
	Total	1090

Table 18: http code status distribution

In the future, the system's 1.7-second response time may undoubtedly be shortened. In a similar vein, the system can manage a load of roughly ten requests per second and has a 4.1% request failure rate. Even while the performance test shows that the backend is generally reliable, scalable, and responsive, there is certainly room for improvement in the near future.

Chapter 10: Critical Analysis

The AR mount guide project was designed utilizing an agile methodology that separated the requirements study, development, and testing phases. This artifact's primary objective is to create a reliable system that enables users to scan their environment and obtain an augmented reality object if a mountain is spotted. The software also has AR mountain scan and regular overlay. Those who wish to scan their surroundings and have 2D text labels shown on the screen are the target audience for normal overlay. Additionally, an AR scan makes use of augmented reality's capabilities to show an AR object on a mountain screen. Similar to this, a 3D model with a thorough description is shown when the mountain is identified. The AR Mount Guide project is critically reviewed in this chapter, with an emphasis on the strength, weakness and possible improvement for future.

The project's goals and objectives come after the earliest stages, which concentrate on identifying the issue and locating an appropriate solution. Additionally, a literature review was created after research was done on a number of journals, papers, and websites. After that, it concentrates on listing features, designing the system, and describing the technology and tools utilized.

After that, there were a number of issues and difficulties encountered during the early stages of development. The artifact was originally intended to be developed for both Android and iOS, however it was challenging to proceed with iOS because Apple devices were not available. Thus, on Android smartphones, the artifact functions flawlessly. The fact that it is limited to Android devices is one of its main drawbacks. Furthermore, there were numerous problems with Flutter's AR integration. First off, since the app is Android-based, ARKit is not required; instead, ARCore is used in this project. Additionally, because Google hasn't maintained the `arcore_flutter_plugin` package for the past two years, there were numerous problems. During the development of AR, there were a lot of flaws, but they were eventually fixed. However, the AR mountain scan will only show five mountains because using many mountains will make the program sluggish and occasionally crash; this is another flaw.

Furthermore, since combining all of the 3D models can be expensive, there will only be one 3D model for every mountain found. In addition, the software needs to identify the

plane in order to properly put AR. The AR won't be positioned correctly otherwise. In a similar vein, the app can take a long time to identify the AR. This is the AR mount guide's flaw as well.

Despite these flaws, the mount guide project is improved by a number of aspects. People can use the standard overlay if the AR is inappropriate for them. Additionally, the project can enhance people's mountain experiences while simultaneously supporting the field of education. Both teachers and students will have a better teaching and learning experience as a result. This allows the student to gain real-time knowledge about mountains. In a similar spirit, the following app will definitely increase tourism because a lot of visitors come to Nepal and would need to use it for efficient mountain viewing instruction.

Furthermore, the software can undoubtedly get better in the future. To begin with, more mountains could be added so that users could explore them in augmented reality. The user experience can undoubtedly be enhanced by this. Additionally, if the software can be developed on iOS as well, it will gain more users. This benefits the application's entire market since more people are using it. Furthermore, the app can undoubtedly gain from the combination of augmented reality and artificial intelligence. In a similar vein, the user experience will be improved if each mountain has its own 3D model.

In general, the process of creating the AR mount guide project involves research, development, and testing. Although the AR mount guidance app isn't flawless, there are undoubtedly opportunities for future enhancements.

Chapter 11: Limitation and Future scope

11.1 Limitation

Therefore, there are several obstacles and technical problems involved in creating a real-time AR application that incorporates Haversine computation, bearing calculation, AR object placement, and label positioning. Throughout the development process, numerous issues arose. A list of this application's restrictions and typical problems may be found below.

1. **Device Compatibility Limitation:** Despite being cross-platform, ARCore only works with Android devices because the entire project is made with the ARCore plugin in Flutter. This implies that this application is inaccessible to iPhone users. Furthermore, Android devices need to have ARCore services installed on their cellphones because it leverages Google ARCore.
2. **Limited mountain overlay:** Thus, only six mountains are used in an AR overlay for smooth functioning because adding too much mountain data causes AR to become slow and also drains the battery. For instance, AR tends to be slow and the app fails right away if ten mountains are detected in a single frame. Thus, the AR mount guide displays a limited number of mountains. Similarly, it can be inefficient to recalculate bearing or distances on each update if there are many mountains.
3. **Sensor Accuracy and Environmental Interference:** The following application mostly uses Flutter Compass for GPS positioning and heading, but it can be inaccurate in areas with poor service. Similarly, AR could not be demonstrated without effective illumination. Therefore, for AR to work, the right lighting and surroundings are needed.
4. **Battery consumption:** In this case, using AR for extended periods of time simultaneously depletes the battery and results in performance optimization. In order to ensure a seamless AR user experience, the AR screen shouldn't be kept open for extended periods of time. However, this can negatively impact the user experience.
5. **Permission Management:** Thus, the application requires a number of permissions. For example, the permissions needed for AR to function correctly are camera and

user location. Therefore, the app won't be able to access the camera and the user won't be able to use the augmented reality application if camera and user location permissions are denied.

6. Ar Overlay speed: Due to the 3D object placement, ar application can be slow and can result in slow processing of ar overlay. Similarly when ar is detected user will be able to view basic sphere in earth texture.

11.2 Future Scope

The program has a great deal of potential to grow into a more scalable and optimized version of the AR mount guide application because it makes use of contemporary technologies like augmented reality.

1. Expansion into cross- platform mobile application: The program can therefore make use of the ARKit plugin, which is appropriate for iOS devices, since it makes use of the ARCore Flutter plugin for Android. These have the potential to reach a larger audience and generate more user feedback. Therefore, the application may be made cross-platform by integrating both ARCore and ARKit.
2. Utilization of 3d mountain model: By using 3D models of various mountains, the software can improve user experience rather than depending solely on sphere 3D shapes with earth texture.
3. Changing the tech stack: Thus, Google ARCore and Flutter are being employed for the application. However, Google neglected to support Google ARCore for two years, which led to a lack of community, documentation, and resources. Here, using Unity in instead of Flutter will aid with both seamless AR overlay and quick 3D model rendering. Therefore, switching to Unity can improve AR Mountain Guide's overall performance.
4. More Mountain Data: Therefore, if we make use of all of Nepal's mountains, they may be applied in the actual world and offer superior feedback. Therefore, expanding the application to a larger version that can make use of more mountains can undoubtedly improve the user experience.

5. Integration of AI: Given the current state of AI, utilizing TensorFlow Lite or PyTorch to detect real mountains from camera frames using picture classification can improve performance.

Chapter 12: Conclusion

To put it briefly, the research study aims to develop a revolutionary mobile-based AR application that can assist individuals to improve their experience in viewing mountains. The app provides users with two simulation modes, such as normal overlay and AR scan overlay. In the AR overlay, the user can scan the surroundings, and when a mountain is detected, an AR object will be displayed on the screen. Similarly, when clicked on the detected mountain, the user can view the 3D model of the detected mountains. Similarly, in the normal overlay, the user can view the 2D label of the detected mountains.

The app presents to offer a full-stack mobile application with responsive and user-intuitive mobile app design, making it both entertaining and instructive. It does this by utilizing Flutter, Node.js, and MongoDB. Furthermore, the software is usable by a wide range of users across various age groups. Although this application has certain drawbacks, including hardware and environmental restrictions, it also provides opportunities for future development. The provided research paper has been a great learning experience by deep diving into the world of augmented reality, geo location, compass and haversine distance calculation.

All things considered, the AR Mount Guide streamlines exploration by simplifying and free of cost mountain identification. This software lays a solid foundation that supports augmented reality applications in both indoor and outdoor simulations. Overall, the app can be utilized by both regular people and tourist where they can enhance their mountain exploration process.

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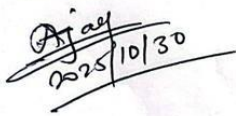
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https://www.thomasweibel.ch/data/uploads/text/mont_en.pdf [Accessed 9 Apr. 2025]

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Chapter 14 : Appendix

14.1 Weekly Progress report for Undergraduate project

 	
UNIVERSITY OF BEDFORDSHIRE DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY FINAL YEAR UG PROJECT WEEKLY PROGRESS REPORT FORM	
Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 13	Report No: 1
Summary of progress (including any problems)	- Gathered essential resources, including 3D models of mountains for visualization. - Collected and organized relevant mountain data for integration. - Developed the system design in alignment with the specified requirements.
Plan for next week	- Initiated the prototype design using Figma. - Began backend development for API implementation. - Tested API endpoints using Postman to ensure functionality and reliability.
Supervisor Comment	- Prototype design using figma - start to design App (configuration)
Student's Signature <i>Minal</i>	Supervisor's Signature <i>Krishna</i>
Date 10 th Oct, 2025	Date 30/10/2025
	
When signed this form must be scanned and submitted via the relevant link on BREO	



UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR UG PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 21	Report No: 2

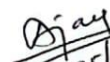
Summary of progress (including any problems)	- Initialized the prototype design in Figma - Start developing API in node.js - Worked out and planned the distance calculation using the Haversine formula
Plan for next week	- Complete the prototype design - Began testing distance calculation accuracy using sample coordinates - Tested API endpoints using Postman to ensure functionality and reliability.
Supervisor Comment	- Prototype - comple - Database Schemadipision - Develop App for

Student's Signature 

Date 12th Oct, 2023

Supervisor's Signature 

Date 30/10/2023


2023/10/30

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UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR UG PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 5	Report No: 3

Summary of progress (including any problems)	- Completed the basic ui prototype design for Ar Mount Guide - Integrated MongoDB schema for mountain information storage - Created backend routes for fetching nearby mountain data based on current geolocation. - Tested API endpoints using Postman to ensure functionality and reliability
Plan for next week	- Start Developing Frontend in flutter - Proper setup using MVVM model - Starting working on AR workflow - Researched on suitable Ar library such as ArCore and ARKit
Supervisor Comment	- Flutter App Configuration - AR code pace - test

Student's Signature *Minal*

Date ... 24/10/2025

Supervisor's Signature *K. Aryal*

Date ... 30/10/2025

Ajay
 25/10/30

When signed this form must be scanned and submitted via the relevant link on BREO



UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR UG PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 46	Report No: 4

Summary of progress (including any problems)	- Start working on both normal and Ar overlay - Implement location tracking in flutter - Starting creating basic shapes in ar using ArCore Package - Implemented mountain marker in ArView
Plan for next week	- Implementing the real time detection of mountain - Improved backend data filtering to return mountain with specific distance - Complete Normal Overlay
Supervisor Comment	- Real time detection - Complete Front-end design

Student's Signature MinalDate 30th Oct., 2025Supervisor's Signature KrishnaDate 30/10/2025

Djay
2025/10/30

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UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR UG PROJECT

WEEKLY PROGRESS REPORT FORM

Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 7	Report No: 5

Summary of progress (including any problems)	- Implementing Authentication - Implementing the real time detection of mountain in normal overlay - Improved Responsiveness of frontend
Plan for next week	- Start working on Ar detection of multiple mountains - Improve better api response if required - Start writing reflective report
Supervisor Comment	✓ Work on reflective report ✓ Integrate Ar Mountain overlay

Student's Signature *Minal*

Supervisor's Signature *Krishna*

Date *7 Nov 2025*

Date *9 Nov 2025*

22/11/25

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UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR UG PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 6	Report No: 6

Summary of progress (including any problems)	- Tested <u>AR Marker</u> in ar view - Refined distance calculation formula to improved accuracy - Implementing caching of mountain data for offline support - Start developing report
Plan for next week	- Start final documentation - Prepare demo test for the application
Supervisor Comment	• Documentation - draft to final • Application - final Test/Bug/Comments - optimize the codes;

Student's Signature MinalSupervisor's Signature KrishnaDate 13th Nov, 2025Date 13th Nov, 2025

Djay
2025/11/13

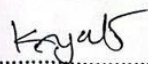
When signed this form must be scanned and submitted via the relevant link on BREO



UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR UG PROJECT
WEEKLY PROGRESS REPORT FORM

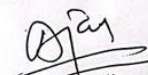
Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 9	Report No: 7

Summary of progress (including any problems)	- Reflective Submission - Start the thesis report - Test the application
Plan for next week	- Work on user evaluation and testing - Continue with the development
Supervisor Comment	• Finalization of documentation • App Testing & Rep case

Student's Signature Supervisor's Signature 

Date Nov 15, 2025

Date Nov 19 2025


20/11/19

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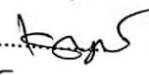


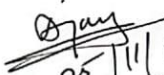
UNIVERSITY OF BEDFORDSHIRE
DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY
FINAL YEAR UG PROJECT
WEEKLY PROGRESS REPORT FORM

Student's Name: Minal Pariyar	Supervisor's Name: Krishna Aryal
Project Title : Ar Mount Guide	
Week No: 10	Report No: 8

Summary of progress (including any problems)	- Tested the application - Fixed few bugs in the application - Continued with the documentation - Write unit test for the application
Plan for next week	- Design the poster - Prepare demo application presentation - Finalise documentation
Supervisor Comment	• finalize the documentation • finalize the application • finalize the presentation & poster

Student's Signature 
Date Start 23rd November

Supervisor's Signature 
Date Nov 27, 2025


2025/11/28

When signed this form must be scanned and submitted via the relevant link on BREO

14.2 Proposal for RMET

BSc Project Screening Form: Guidelines

Part 1 – Project Proposal

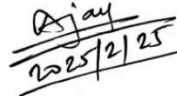
Student Name	Minal Pariyar
Student Number	2214714
Degree Pathway	Bsc (Hons) Computer Science & Software Engineering
Supervisor Name	Krishna Aryal
Title of the Project	AR Mount Guide
Abstract of the project	The following research paper focuses on enhancing mountain exploration for tourists and regular people by utilizing modern technology such as Augmented Reality (AR). AR is gaining more popularity due its immersive user experience where user can view any objects in real scenario. With tourism increasing in Nepal, mountain exploration can be quite difficult for tourists due to its harsh climatic conditions and lack of knowledge regarding surrounding landscapes. Conventional methods, including books and internet searches, might be challenging to learn about the nearby mountains. In order to address that issue, the following research focuses on how the AR Mount Guide project is a revolutionary interactive mobile application that utilizes augmented reality features to enhance user experience in their way of viewing mountains. The study also focuses on determining user's orientation in real time using geolocation, gyroscope and compass data. The project aims to provide an interactive AR experience by allowing users to scan the surroundings and get the mountains located in that particular direction along with their detailed information. Keywords: Mountain Identification, Augmented Reality, AR Tourism, Geospatial tech, AR with Geospatial tech, 3D Model with AR
Project Deliverables	• Fully functional Mobile Application • Indoor Ar Simulation • Contextual Report

	• Final Report • Poster
Description of your artefact	Project Background In Nepal, there is an increasing boost in tourism, with everyday tourists visiting Nepal for trekking and hiking purposes. In spite of the fact that many tourists are visiting Nepal, it is not ideal for every tourist to have knowledge regarding mountains, landscapes, and other geographical features. Even normal citizens donot have full knowledge regarding the geographical features of Nepals. To add to that, people often need to rely on guided books, maps, and any search engine. While these conventional methods are necessary, they do not provide a clear, intuitive, and interactive experience to the user. (Prapas, 2023) In a similar manner augmented reality is becoming more and more popular over time it combines virtual objects with real world within user eye which helps in enhancing the overall user experience (Syed, 2023). Moreover, there are a lot of benefits of using modern technologies such as augmented reality in mobile apps, as it allows users to visualise any objects that make the user experience better (Davidaviciene, 2021). Furthermore, when augmented reality is used along with geospatial technology, it allows users to get real-time interactive data and information regarding any geographical features, which improves tourism and navigation (Manggopa, 2022). Further using Augmented reality with GPS, it helps to enhance overall experience of travelling in the geographical location (Xueqiu Zhuang, 2022). Problem Statement In the present scenario, people often rely on conventional methods such as tour guides, guided books, or any online searches to know about specific mountains. While these methods are necessary, they require a lot of manual effort, where it may be time-consuming, and in some cases, the provided information is not clear and intuitive (Ricart, 2023). Since Nepal is a

	naturally gifted country with many snow-capped mountain peaks, it is difficult for every individual to know about particular peaks. Aim The primary aim of this study is to research on how modern technology like Augmented Reality (AR) can enhance mountain exploration for tourists and for regular people. The objectives of this project are as follows: • To research on Augmented reality • To analyze on compatibility of AR in mobile application • To understand the geospatial technology • To study on geolocation, gyroscope and compass data for real time user data Following are the features that the artefact will include: • AR based real time mountain identification • Users can view 3D model of mountain which is detected. • User can view detailed description of mountains which is detected • Indoor AR mountain simulation																								
Risk analysis	<table><tr><th>Risk</th><th>Impact</th><th>Likelihood</th><th>Mitigation Strategy</th></tr><tr><td>Inaccurate orientation and placement</td><td>High</td><td>Possible</td><td>Implement alternative location detection method</td></tr><tr><td>Improper identification of mountain</td><td>Medium</td><td>Likely</td><td>Enhance recognition accuracy through calibration</td></tr><tr><td>High battery consumption</td><td>High</td><td>Moderate</td><td>Implement power saving techniques for proper optimization</td></tr><tr><td>Complexity in indoor AR simulation</td><td>Medium</td><td>Moderate</td><td>Improve algorithm for AR</td></tr><tr><td>Failure of AR due to poor lighting</td><td>High</td><td>Likely</td><td>Use additional sensor for better accuracy and upgrade AR algorithm</td></tr></table>	Risk	Impact	Likelihood	Mitigation Strategy	Inaccurate orientation and placement	High	Possible	Implement alternative location detection method	Improper identification of mountain	Medium	Likely	Enhance recognition accuracy through calibration	High battery consumption	High	Moderate	Implement power saving techniques for proper optimization	Complexity in indoor AR simulation	Medium	Moderate	Improve algorithm for AR	Failure of AR due to poor lighting	High	Likely	Use additional sensor for better accuracy and upgrade AR algorithm
Risk	Impact	Likelihood	Mitigation Strategy																						
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High battery consumption	High	Moderate	Implement power saving techniques for proper optimization																						
Complexity in indoor AR simulation	Medium	Moderate	Improve algorithm for AR																						
Failure of AR due to poor lighting	High	Likely	Use additional sensor for better accuracy and upgrade AR algorithm																						

Resources Required	Hardware Requirements • Mobile devices with AR compatibility	
	Software Requirements • Android Studio • Git and Github • Vs Code • Mongo Db Atlas	
	Framework and Languages Used • Flutter for frontend • Node js for backend • Mongo Db as NOSQL database • Javascript (backend language) • Dart (frontend language)	
Have you completed & submitted your ethics form?	YES	NO
If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence		

Supervisor Signature:

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

1. Books

- i. Andrew Yeh Ching Nee, Soh Khim Ong (2023) Springer Handbook of Augmented Reality
https://books.google.com.np/books?id=hdakEAAQBAJ&pg=PR5&redir_esc=y#v=onepage&q&f=false

- ii. Andrey Chernov (2022) Augmented Reality in learning Geography landscapes
<https://www.mdpi.com/2571-5577/5/6/111#>

2. Journal Papers

- i. Shaveta Dargan, Shally Bansal, Munish Kumar, Ajay Mittal, Krishan Kumar (2023) Augmented Reality : A comprehensive Review
<https://link.springer.com/article/10.1007/s11831-022-09831-7>
- ii. Calin Gheorghe Dan Neamtu, Radu Comes, Jing-Jing Fang (2022) In Depth Review of Augmented Reality <https://www.mdpi.com/1424-8220/23/1/146>
- iii. Tai-Yi Yu, Nae-Wen Kuo, Mei-Lan Lin (2023) Augmented Reality and Toursim
<https://www.mdpi.com/2071-1050/15/21/15358>
- iv. Bharath G.; Rupa Ch; Karthik M.; Leela Bhavani A.; Manish Chowdary V. (2021) Revelation of Geospatial Information using Augmented Reality
<https://ieeexplore.ieee.org/abstract/document/9419459>

3. Relevant Software

- i. ArCore : <https://www.jstor.org/stable/1791850>
- ii. Android Studio Installation :
https://developer.android.com/studio?gad_source=1&gclid=CjwKCAiAzvC9BhADEiwAEhtlN-aUuwQxWhZMgHTTTPBLTl0zXt39VOiLXlcljWT3cMQ_xlhKXq5BDuBoCpFoQAvD_BwE&gclidsrc=aw.ds
- iii. Blender : <https://www.blender.org/>
- iv. Node js : <https://nodejs.org/en/download>
- v. Mongo DB: <https://www.mongodb.com/>

4. Relevant Hardware

- i. Mobile device with AR compatibility :
<https://ieeexplore.ieee.org/abstract/document/9363323>

5. Others

- a) Integration of AR with Geospatial visualization :
<https://www.researchgate.net/directory/publications>
- b) History of Augmented Reality :
https://ronaldazuma.com/papers/CG_EditorIntro_2011.pdf

14.3 Proposal for RMET

FACULTY OF CREATIVE ARTS, TECHNOLOGIES AND SCIENCE**Form for Research Ethics Projects (Ethics Form)**

Student Name	Minal Pariyar
Student Number	2214714
Degree Pathway	Bsc (Hons) Computer Science & Software Engineering
Supervisor name	Krishna Aryal
Supervisor Signature	
Title of project	AR Mount Guide

SECTION A

All the data of mountains were taken from freely available source such as Wikipidea and from Nepal Himal Peak Profile. These data will only be used for research purposes.

SECTION B Check List

Please answer the following questions by circling YES or NO as appropriate.

Does the study involve vulnerable participants or those unable to give informed consent (e.g. children, people with learning disabilities, your own students)?	YES ☒ NO
Will the study require permission of a gatekeeper for access to participants (e.g. schools, self-help groups, residential homes)?	YES ☒ NO
Will it be necessary for participants to be involved without consent (e.g. covert observation in non-public places)?	YES ☒ NO
Will the study involve sensitive topics (e.g. obtaining information about sexual activity, substance abuse)?	YES ☒ NO
Will blood, tissue samples or any other substances be taken from participants?	YES ☒ NO
Will the research involve intrusive interventions (e.g. the administration of drugs, hypnosis, physical exercise)?	YES ☒ NO
Will financial or other inducements be offered to participants (except reasonable expenses or small tokens of appreciation)?	YES ☒ NO
Will the research investigate any aspect of illegal activity (e.g. drugs, crime, underage alcohol consumption or sexual activity)?	YES ☒ NO
Will participants be stressed beyond what is considered normal for them?	YES ☒ NO
Will the study involve participants from the NHS (patients or staff) or will data be obtained from NHS premises?	YES ☒ NO

If the answer to any of the questions above is "Yes", or if there are any other significant ethical issues, then further ethical consideration is required. Please document carefully how these issues will be addressed.

Signed (student):

Date:

Minal 25th Feb, 2025

Countersigned (Supervisor):

Date:

Kyule
25 Feb 2025

14.4 Testing Figures

```
// REGISTER
it("should return 400 if fields are missing during register", async () => {
  const res = await request(app)
    .post("/api/users/register")
    .send({ email: "", username: "", password: "" });

  expect(res.statusCode).toBe(400);
  expect(res.body.message).toBe("All fields are required");
});
```

Figure 32: Missing fields during register

```
it("should return 400 if username is missing during register", async () => {
  const res = await request(app)
    .post("/api/users/register")
    .send({ email: "test@example.com", password: "12345678" });

  expect(res.statusCode).toBe(400);
  expect(res.body.message).toBe("All fields are required");
});
```

Figure 33: Missing username during register

```
it("should return 400 if email is missing during register", async () => {
  const res = await request(app)
    .post("/api/users/register")
    .send({ username: "test", password: "12345678" });

  expect(res.statusCode).toBe(400);
  expect(res.body.message).toBe("All fields are required");
});
```

Figure 34: Missing email during register

```
it("should return 400 if password is missing during register", async () => {
  const res = await request(app)
    .post("/api/users/register")
    .send({ username: "test", email: "test@example.com" });

  expect(res.statusCode).toBe(400);
  expect(res.body.message).toBe("All fields are required");
});
```

Figure 35: Missing password during register

```
it("should return 400 if password is less than 8 characters during register", async () =
  const res = await request(app)
    .post("/api/users/register")
    .send({ username: "test", email: "test@example.com", password: "1234567" });

  expect(res.statusCode).toBe(400);
  expect(res.body.message).toBe("Password must be at least 8 characters");
});
```

Figure 36: Password length

```
it("should return 409 if user already exists", async () => {
  mockingoose(User).toReturn(
    { _id: "507f191e810c19729de860ea", email: "test@example.com" },
    "findOne"
  );

  const res = await request(app)
    .post("/api/users/register")
    .send({
      email: "test@example.com",
      username: "test",
      password: "12345678",
    });

  expect(res.statusCode).toBe(409);
  expect(res.body.message).toBe(
    "User already exists with this email or username"
  );
});
```

Figure 37: User Already exists

```
it("should return 400 if email is missing during login", async () => {
  const res = await request(app)
    .post("/api/users/login")
    .send({ password: "12345678" });

  expect(res.statusCode).toBe(400);
  expect(res.body.message).toBe("Email and password are required");
});
```

Figure 38: missing email during email

```
it("should return 400 if password is missing during login", async () => {
  const res = await request(app)
    .post("/api/users/login")
    .send({ email: "test@example.com" });

  expect(res.statusCode).toBe(400);
  expect(res.body.message).toBe("Email and password are required");
});
```

Figure 39: missing password during login

```
it("should return 401 for invalid credentials (user not found)", async () => {
  mockingoose(User).toReturn(null, "findOne");

  const res = await request(app)
    .post("/api/users/login")
    .send({
      email: "wrong@example.com",
      password: "wrongpassword",
    });

  expect(res.statusCode).toBe(401);
  expect(res.body.message).toBe("Invalid credentials");
});
```

Figure 40: invalid credentials

```
it("should return 401 for incorrect password", async () => {
  const fakeUser = {
    _id: "507f191e810c19729de860ea",
    email: "test@example.com",
    password: "hashed",
  };

  mockingoose(User).toReturn(fakeUser, "findOne");

  jest.spyOn(User.prototype, "checkPassword").mockResolvedValue(false);

  const res = await request(app)
    .post("/api/users/login")
    .send({
      email: "test@example.com",
      password: "wrongpassword",
    });

  expect(res.statusCode).toBe(401);
  expect(res.body.message).toBe("Invalid credentials");
});
```

Figure 41: incorrect password

```
it("should login successfully", async () => {
  const fakeUser = {
    _id: "507f191e810c19729de860ea",
    email: "test@example.com",
    password: "hashed",
  };

  mockongoose(User).toReturn(fakeUser, "findOne");
  mockmongoose(User).toReturn(fakeUser, "findById");

  jest.spyOn(User.prototype, "checkPassword").mockResolvedValue(true);
  jest.spyOn(User.prototype, "generateAccessToken").mockReturnValue("ACCESS_TOKEN");
  jest.spyOn(User.prototype, "generateRefreshToken").mockReturnValue("REFRESH_TOKEN");
  jest.spyOn(User.prototype, "save").mockResolvedValue(undefined);

  const res = await request(app)
    .post("/api/users/login")
    .send({
      email: "test@example.com",
      password: "123456",
    });

  expect(res.statusCode).toBe(200);
  expect(res.body.data.accessToken).toBe("ACCESS_TOKEN");
  expect(res.body.data.refreshToken).toBe("REFRESH_TOKEN");
});
```

Figure 42: successfull login

```
test('normalizeDeg', () {
  expect(normalizeDeg(0), 0);
  expect(normalizeDeg(180), -180);
  expect(normalizeDeg(450), 90);
  print("✓ normalizeDeg Test Passed");
});
```

Figure 43: normalise deg test

```
test('bearingDeg', () {
  final brng = bearingDeg(51.5074, -0.1278, 40.7128, -74.0060);
  expect(brng, closeTo(285, 5));
  print("✓ bearingDeg Test Passed");
});
```

Figure 44: bearing test

```
test('haversineKm', () {  
  final d = haversineKm(51.5074, -0.1278, 48.8566, 2.3522);  
  expect(d, closeTo(343, 10));  
  print("✓ haversineKm Test Passed");  
});
```

Figure 45: haversine test

```
test('angularDiffDeg', () {  
  expect(angularDiffDeg(350, 0), 10);  
  print("✓ angularDiffDeg Test Passed");  
});  
);
```

Figure 46: angular diff deg test

```
test('calculateARPosition', () {  
  final pos = calculateARPosition(0, 0, 0);  
  expect(pos.z, closeTo(-1.5, 0.01));  
  print("✓ calculateARPosition Test Passed");  
});  
;
```

Figure 47: calculate ar position test

```
test('getMountainColor', () {  
  final c1 = getMountainColor('Mount Everest');  
  final c2 = getMountainColor('Mount Everest');  
  expect(c1, c2);  
  print("✓ getMountainColor Test Passed");  
});  
;
```

Figure 48: get mountain color test

```
test('makeTextPng', () async {  
  final bytes = await makeTextPng(  
    text: "Test",  
    bgColor: Colors.black,  
    textColor: Colors.white,  
  );  
  
  expect(bytes.length, greaterThan(1000));  
  print("✓ makeTextPng Test Passed");  
});
```

Figure 49: make test png

```
test('getVisibleMountains', () {  
  final mock = MockArScanState();  
  mock.currentAzimuthDeg = 0;  
  mock.mountains = [  
    const Mountain(  
      name: "Mount Everest",  
      latitude: 0,  
      longitude: 0,  
      altitude: 0,  
      country: "",  
      region: "",  
      range: "",  
      difficulty: "",  
      firstAscent: "",  
      description: "",  
    ),  
  ];  
  
  mock.mountainBearings = {'Mount Everest': 10};  
  
  expect(mock.getVisibleMountains(), ['Mount Everest']);  
  print("✓ getVisibleMountains Test Passed");  
});
```

Figure 50: get visible mountains

```
test('calculateMountainData', () {  
  final calc = MockCalculator();  
  
  final mountains = [  
    const Mountain(  
      name: "Mount Everest",  
      latitude: 27.9881,  
      longitude: 86.9250,  
      altitude: 8848,  
      country: "",  
      region: "",  
      range: "",  
      difficulty: "",  
      firstAscent: "",  
      description: "",  
    ),  
  ];  
  
  calc.calculateMountainData(27.0, 86.0, mountains);  
  
  expect(calc.mountainBearings['Mount Everest'], isNotNull);  
  print("✓ calculateMountainData Test Passed");  
});
```

Figure 51: calculate mountain data

14.5 Google form questionnaire of RMET

AR Mount Guide

B *I* U  

Hey there! I'm **Minal Pariyar**, and I'm creating an exciting **AR Mount Guide** app that helps you **identify mountains in real time** using **Augmented Reality (AR)**! Whether you're a nature lover, a casual explorer, or someone who just enjoys beautiful views, I'd love your thoughts on this project.

Your input will help shape this app into the ultimate trekking companion! Ready? Let's go! 🔥🔥

What's your name?

Short answer text

How old are you? (No worries, you're always young at heart!)

☐ 18-25 🧑🏿 Young explorer!

☐ 26-35 🧑🏿 Adventure pro!

☐ 36-45 🧑🏿 Trailblazer!

☐ 46+ 🧑🏿 Mountain master!

How often do you go on nature adventures?

☐ Almost every day 🧑🏿

☐ A few times a month 🧑🏿

☐ A few times a year 🧑🏿

☐ Rarely, but I love looking at pictures! 🧑🏿

Have you ever been to a mountain or hilly area?

☐ Yes – I've conquered a few peaks!

☐ No – But I dream of it!

Do you find it easy to recognize mountain names?

☐ Yes – I've conquered a few peaks!

☐ No – But I dream of it!

☐ Somewhat – I know a few famous ones!

Tt

When you see a breathtaking mountain, how do you find out more about it?

☐ 📱 Google it! (Instant knowledge, no trekking required)

☐ 🗣️ Ask a friend or tour guide (Word of mouth never fails!)

☐ 📖 Read books or travel guides (Old school but gold!)

☐ 🌄 I just admire the view! (No questions, just vibes 🥰)

Ever wished for an easier way to instantly know the name of a mountain?

☐ OMG, YES!! – Where has this been all my life?!

☐ Yes – That would be a game-changer! No – I enjoy the mystery!

☐ No – I enjoy the mystery!

Have you ever used an AR app before?

☐ Yes – I'm an AR pro!

☐ No – But I'm excited to try!

Have you ever used an AR app before?

☐ Yes – I'm an AR pro!

☐ No – But I'm excited to try!

What features would make this AR Mountain Guide awesome? (Rank them!)

☐ 🏔️ Real-time mountain identification

☐ 🏠 3D mountain views

☐ 🌐 Offline mode (because WiFi doesn't hike with you!)

☐ 📖 Fun facts & history about mountains

☐ 🗺️ Navigation assistance

☐ 🏠 Indoor AR mountain simulation

Would you love to see mountain history & fun facts pop up in AR?

☐ Yes – I love learning while exploring!

☐ No – Just show me the mountains!

Would you love to see mountain history & fun facts pop up in AR?

- ☐ Yes – I love learning while exploring!
- ☐ No – Just show me the mountains!

If you could use AR to explore any natural wonder, which one would you choose?

- ☐ Mount Everest – The king of mountains!
- ☐ Mount Kangchenjunga – The mystical giant!
- ☐ Annapurna – Beauty and danger in one!
- ☐ Mount Makalu – The underdog of the Himalayas!

Have you ever tried an AR app before?

- ☐ Yes, and I love AR!
- ☐ Yes, but only briefly
- ☐ No, but I'd love to try!

What would make this app a must-have for you?

- ☐ Super easy to use
- ☐ Accurate mountain identification
- ☐ Fun & interactive experience
- ☐ Lightweight & fast
- ☐ Works offline

Want to be a beta tester and try it first?

- ☐ Yes – I'd love to be part of this! 🙌
- ☐ No – I'll wait for the grand reveal!

Any fun ideas or crazy suggestions?(Let your imagination run wild)

Long answer text

Form Responses1				
Timestamp	What's your name?	How old are you? (No worries, you're always)	How often do you go on nature adventures?	Have you ever been to a mountain or hilly an
3/26/2025 11:00:25	Who cares	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:26	Who cares	26-35 🧡 Adventure pro!	Rarely, but I love looking at pictures! 📷	No – But I dream of it!
3/26/2025 11:00:26	Bloody hell	36-45 🧡 Trailblazer!	A few times a month 🌲	No – But I dream of it!
3/26/2025 11:00:27	Who cares	36-45 🧡 Trailblazer!	Almost every day 🌲	No – But I dream of it!
3/26/2025 11:00:28	Bloody hell	18-25 🧡 Young explorer!	Almost every day 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:29	Damn it	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:30	Bloody hell	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:31	Damn it	26-35 🧡 Adventure pro!	A few times a month 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:32	Bloody hell	18-25 🧡 Young explorer!	A few times a year 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:33	Bloody hell	26-35 🧡 Adventure pro!	A few times a year 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:34	Bloody hell	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:34	Who cares	36-45 🧡 Trailblazer!	Almost every day 🌲	No – But I dream of it!
3/26/2025 11:00:36	Who cares	18-25 🧡 Young explorer!	A few times a month 🌲	No – But I dream of it!
3/26/2025 11:00:37	Damn it	36-45 🧡 Trailblazer!	Rarely, but I love looking at pictures! 📷	Yes – I've conquered a few peaks!
3/26/2025 11:00:38	Damn it	18-25 🧡 Young explorer!	A few times a month 🌲	No – But I dream of it!
3/26/2025 11:00:38	Bloody hell	18-25 🧡 Young explorer!	A few times a year 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:40	Interesting	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:41	Who cares	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:42	Damn it	36-45 🧡 Trailblazer!	A few times a month 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:42	Bloody hell	18-25 🧡 Young explorer!	A few times a month 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:44	Who cares	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:45	Bloody hell	18-25 🧡 Young explorer!	A few times a year 🌲	Yes – I've conquered a few peaks!
3/26/2025 11:00:46	Bloody hell	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:46	Who cares	26-35 🧡 Adventure pro!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:47	Interesting	18-25 🧡 Young explorer!	A few times a year 🌲	No – But I dream of it!
3/26/2025 11:00:48	Bloody hell	36-45 🧡 Trailblazer!	A few times a month 🌲	No – But I dream of it!

Do you find it easy to recognize mountain nar	When you see a breathtaking mountain, how	Ever wished for an easier way to instantly kno
No – But I dream of it!	🧡 Ask a friend or tour guide (Word of mouth nev	Yes – That would be a game-changer! No – I e
Somewhat – I know a few famous ones!	🌲 I just admire the view! (No questions, just vibe	OMG, YES!! – Where has this been all my life?!
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No – But I dream of it!	🧑🏫 Google it! (Instant knowledge, no trekking requ	Yes – That would be a game-changer! No – I e
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Have you ever used an AR app before? ▾	What features would make this AR Mountain ▾	Would you love to see mountain history & fu ▾	If you could use AR to explore any natural w ▾
Yes – I'm an AR pro!	3D mountain views	Yes – I love learning while exploring!	Mount Makalu – The underdog of the Himalayas!
No – But I'm excited to try!	Indoor AR mountain simulation	No – Just show me the mountains!	Mount Makalu – The underdog of the Himalayas!
No – But I'm excited to try!	Real-time mountain identification	No – Just show me the mountains!	Mount Everest – The king of mountains!
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If you could use AR to explore any natural w ▾	Have you ever tried an AR app before? ▾	What would make this app a must-have for ▾	Want to be a beta tester and try it first? ▾	Any fun ideas or crazy suggestions?(Let your ▾
Mount Makalu – The underdog of the Himalayas!	Yes, but only briefly	Accurate mountain identification	Yes – I'd love to be part of this! 🙌	Who cares
Mount Makalu – The underdog of the Himalayas!	Yes, but only briefly	Lightweight & fast	No – I'll wait for the grand reveal!	Interesting
Mount Everest – The king of mountains!	Yes, but only briefly	Fun & interactive experience	No – I'll wait for the grand reveal!	Bloody hell
Mount Everest – The king of mountains!	Yes, and I love AR!	Super easy to use	Yes – I'd love to be part of this! 🙌	Who cares
Mount Everest – The king of mountains!	Yes, and I love AR!	Works offline	No – I'll wait for the grand reveal!	Damn it
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14.6 Poster